AI VIET NAM
@aivietnam.edu.vn

Statistic and Its Application
(Covariance & Correlation Coefficient)

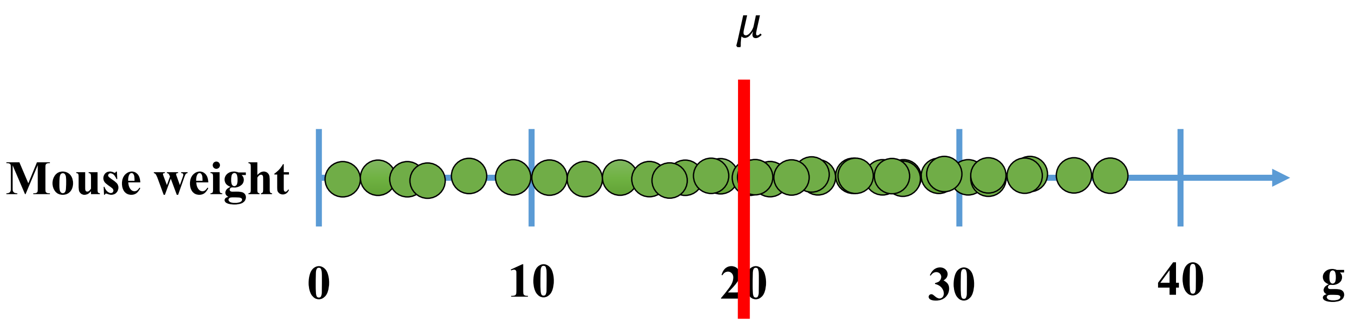
Vinh Dinh Nguyen
PhD in Computer Science

Outline

- Mean, Variance and Standard Deviation: Review
- What is Correlation?
- What is Covariance?
- What is Correlation Coefficient
- Template Matching
- Summary

AI VIET NAM
@aivietnam.edu.vn

Population Mean



Calculate Population mean: $\mu = \frac{1}{n}\sum_{k=1}^{n} x_i$

Calculate variance $var(X) = \frac{1}{n}\sum_{k=1}^{n} (x_i - \mu)^2$

Calculate Standard deviation $\sigma = \sqrt{var(X)}$

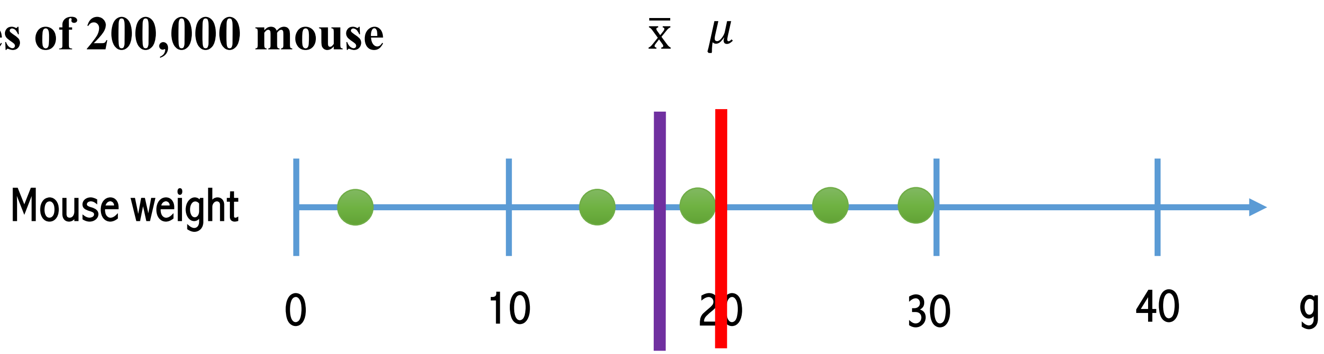
Calculate Population mean: $\frac{1+3+5+\cdots+35}{200,0000}$

Rarely have enough time and money to collect and measure Data → Estimate the Population Mean by using a small sample

AI VIET NAM
@aivietnam.edu.vn

Sample Mean

5 samples of 200,000 mouse



Estimate population mean: $\bar{x} = \frac{1}{n}\sum_{k=1}^{n} x_i$

Estimate variance $var(X) = \frac{1}{n}\sum_{k=1}^{n}(x_i - \bar{x})^2$

Estimate Standard deviation $\sigma = \sqrt{var(X)}$

Sample Mean ($\bar{x}$): $\frac{3+13+19+24+29}{5} = 17.6$

Estimate Population Mean

Problem: We would consistenly underestimate the variance around the population mean. Why?

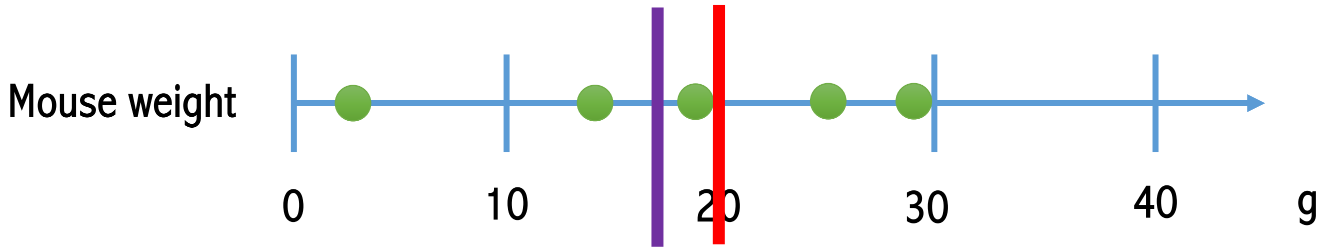
Why diving by n underestimate the population variance?

Sample Mean

5 samples of 200,000 mouse

$\bar{x} \quad \mu = 20$

$$\frac{1}{n}\sum_{k=1}^{n}(x_i - z)^2$$



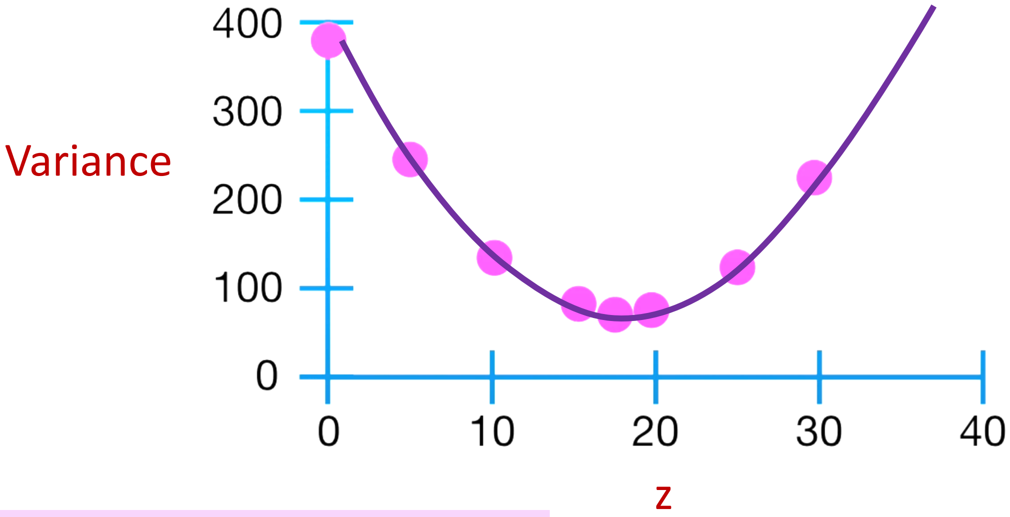
Find the mean z that minimizes the variance

$$\frac{d\frac{1}{n}\sum(x-z)^2}{dz} = 0 \;\Rightarrow\; \frac{\sum 2(x-z)\times -1}{n} = 0 \;\Rightarrow\; -\frac{2}{n}\sum 2(x-z) = 0$$

-2/5 * [(3-z) + (13-z) + (19-z) + (24-z) + (29-z)] = 0

z = (3 + 13 + 19 + 24 + 29)/5

This is the sample mean

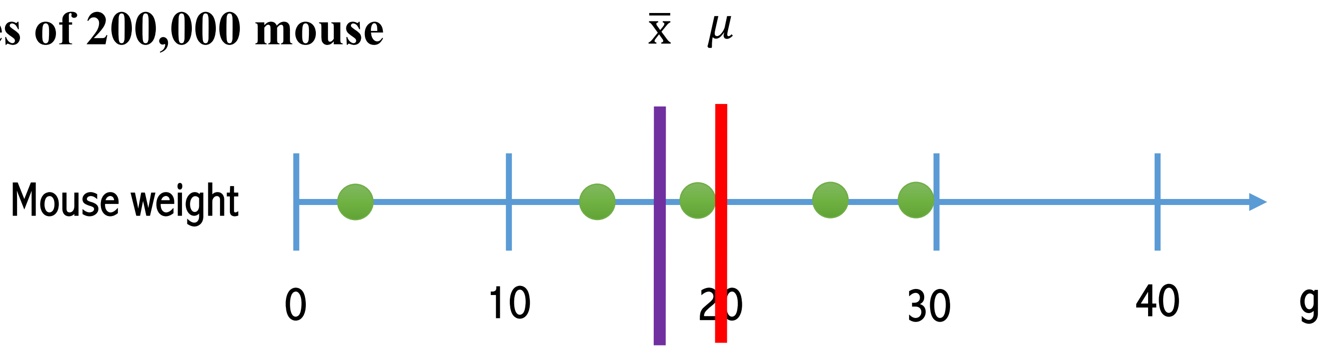


The value around the sample mean is always less than the value around the population mean

$$\frac{\sum(x-\bar{x})^2}{n} = 81.44 \quad < \quad \frac{\sum(x-\mu)^2}{n} = 87.2$$

Sample Mean

5 samples of 200,000 mouse



Population

Calculate Population mean: $\mu = \frac{1}{n}\sum_{k=1}^{n} x_i$

Calculate variance $var(X) = \frac{1}{n}\sum_{k=1}^{n} (x_i - \mu)^2$

Calculate Standard deviation $\sigma = \sqrt{var(X)}$

Sample

Estimate population mean: $\bar{x} = \frac{1}{n}\sum_{k=1}^{n} x_i$

Estimate variance $var(X) = \frac{1}{n-1}\sum_{k=1}^{n} (x_i - \bar{x})^2$

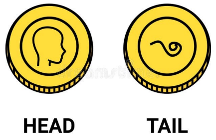
Estimate Standard deviation $\sigma = \sqrt{var(X)}$

Why not n-2, n-3,…?

Degrees of Freedom

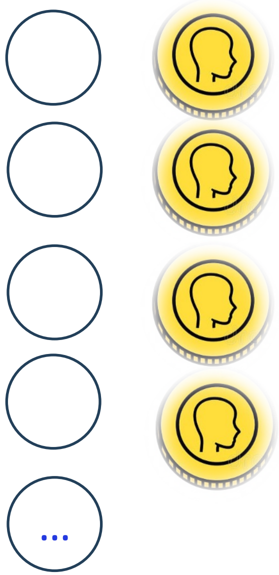
In [statistics](), the number of **degrees of freedom** is the number of values in the final calculation of a [statistic]() that are free to vary (from Wikipedia)

I Tossed a Coin by 100 times:
You want to know how many heads and tails I got



Two possible outcomes

1 degree of freedom



You just need to ask me 1 question (information):
how many heads that I got?

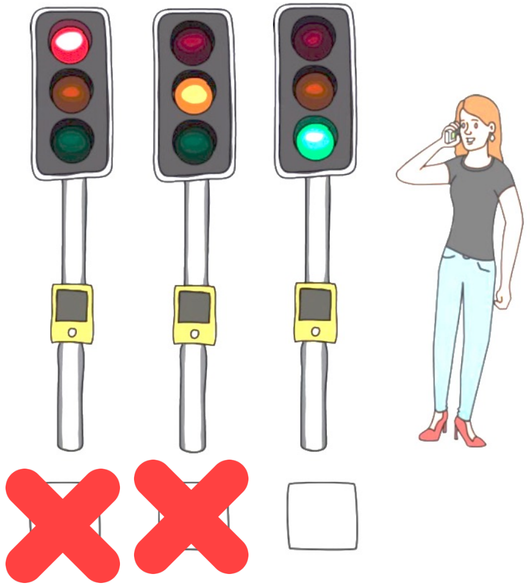
Then, If you know the number of heads I got => you automatically know the number of tails

Degrees of Freedom

In statistics, the number of **degrees of freedom** is the number of values in the final calculation of a statistic that are free to vary (from Wikipedia)

You are on a call and ask your friend about the color of the traffic light.

It is neither yellow nor red.



Two pices of information required to know the color of the light

2 degree of freedom

3 possile outcomes: Red, Yellow, and Green

Sample Mean

In statistics, the number of **degrees of freedom** is the number of values in the final calculation of a statistic that are free to vary (from Wikipedia)

If you know the mean, the last value in your sample is no longer independent

If you know the mean, the last value in your sample is no longer independent

$\bar{x} = 5$

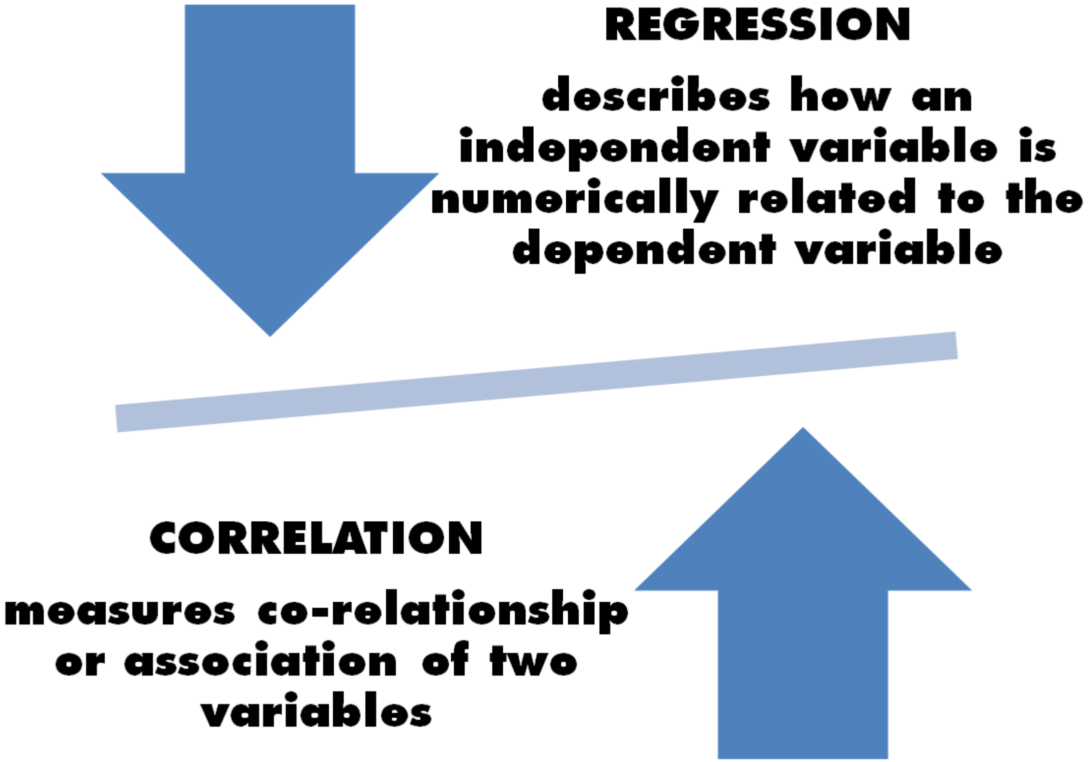
4 | 5 | ?

Estimate variance $var(X) = \frac{1}{n-1}\sum_{k=1}^{n}(x_i - \bar{x})^2$

Outline

- ➢ Mean, Variance and Standard Deviation: Review
- ➢ What is Correlation?
- ➢ What is Covariance?
- ➢ What is Correlation Coefficient
- ➢ Template Matching
- ➢ Summary

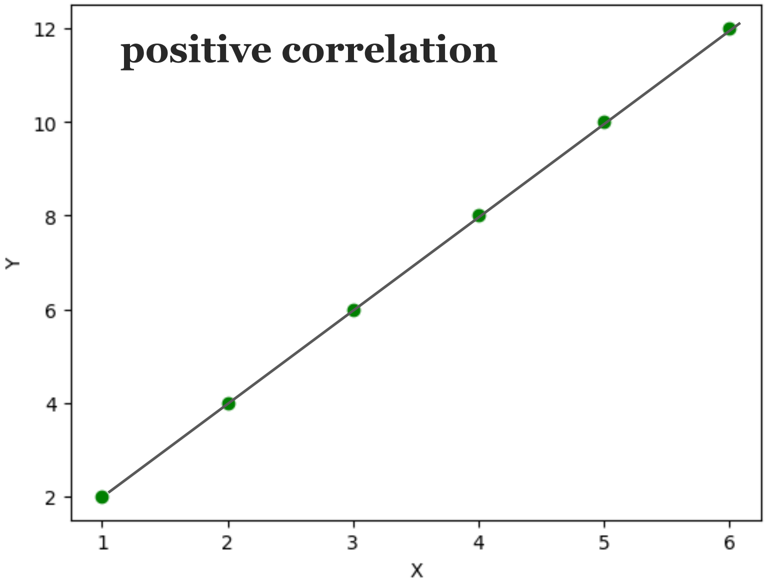
Correlation vs Regression



The key **difference** between correlation and regression is that correlation measures the degree of a relationship between two independent variables (x and y). In contrast, regression is how one variable affects another.

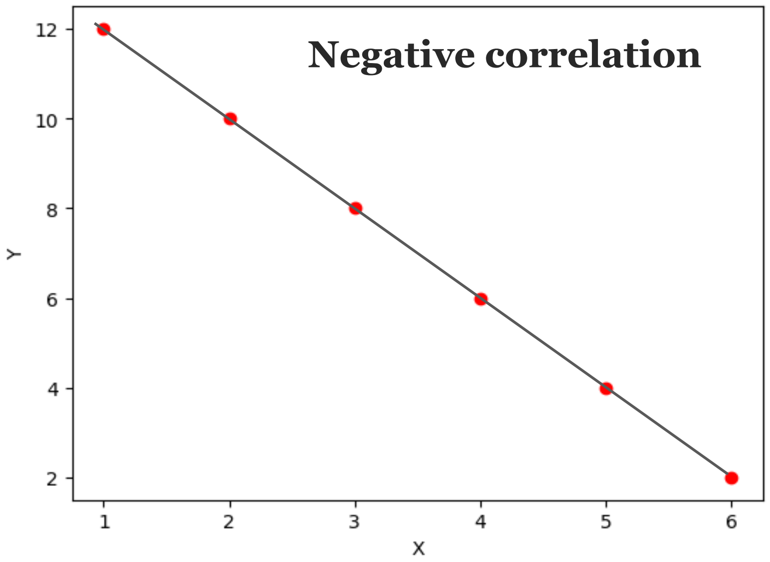
Correlation Between Two Variables

X	1	2	3	4	5	6
Y	2	4	6	8	10	12



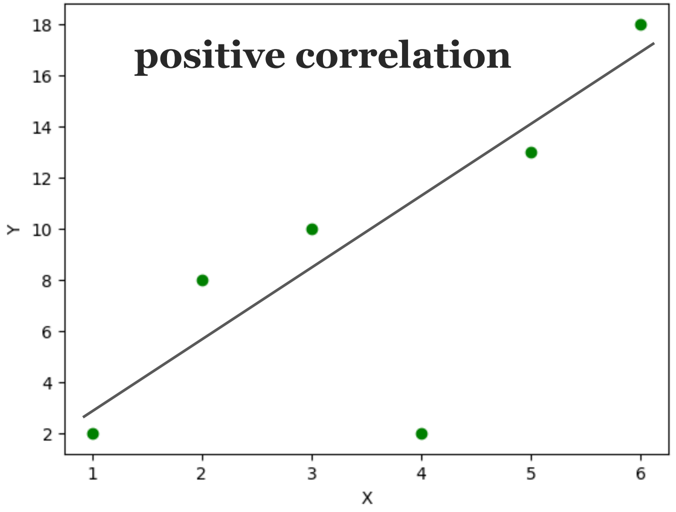
Correlation ~ + 1
(X increase, Y Increase)
Positive Slope
Point on the Line

X	1	2	3	4	5	6
Y	12	10	8	6	4	2



Correlation ~ -1
(X increase, Y Decrease)
Negative Slope
Point on the Line

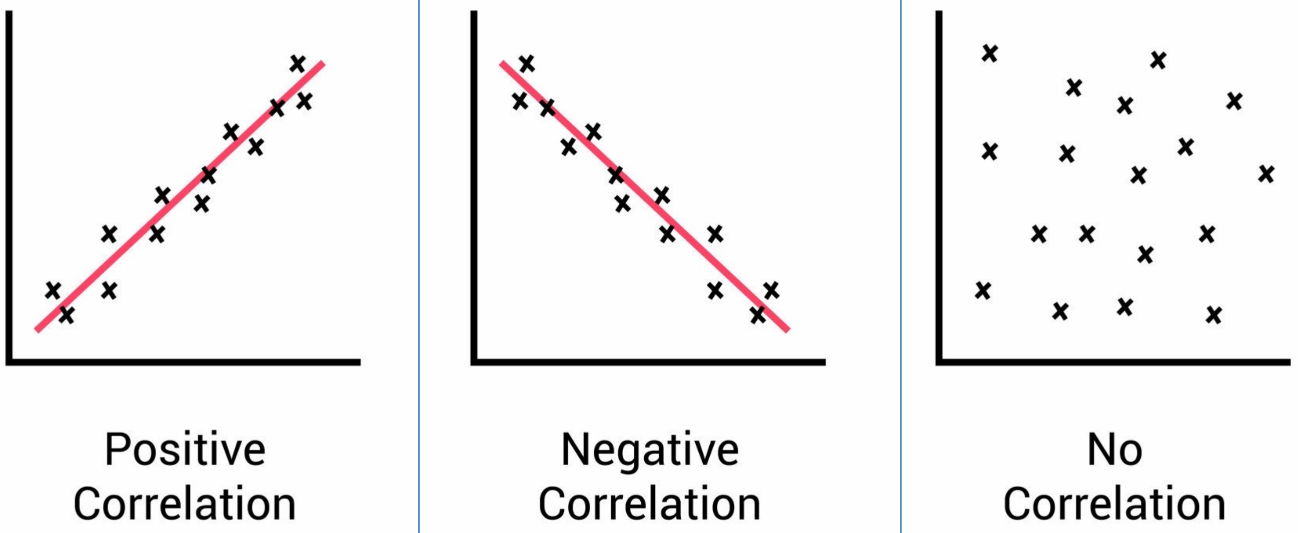
X	1	2	3	4	5	6
Y	2	6	7	9	13	14



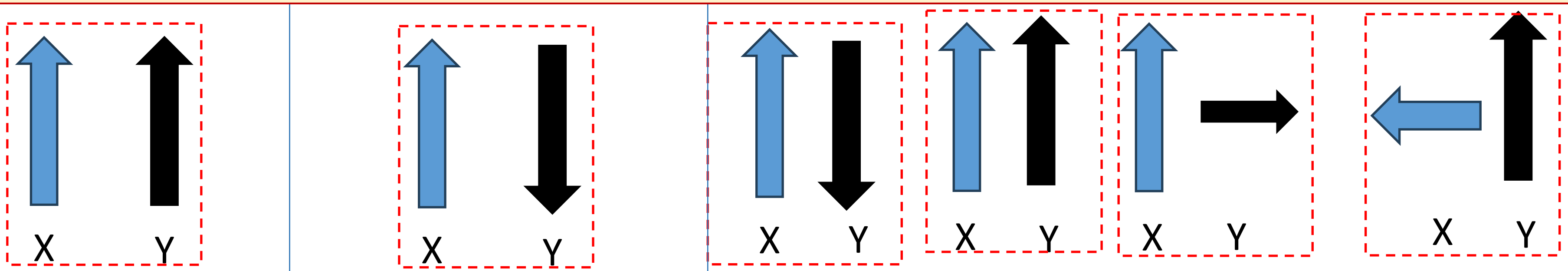
0 < Correlation < 1
(X increase, Y Increase)
Positive Slope
Point not on the Line

AI VIET NAM
@aivietnam.edu.vn

Correlation Between Two Variables



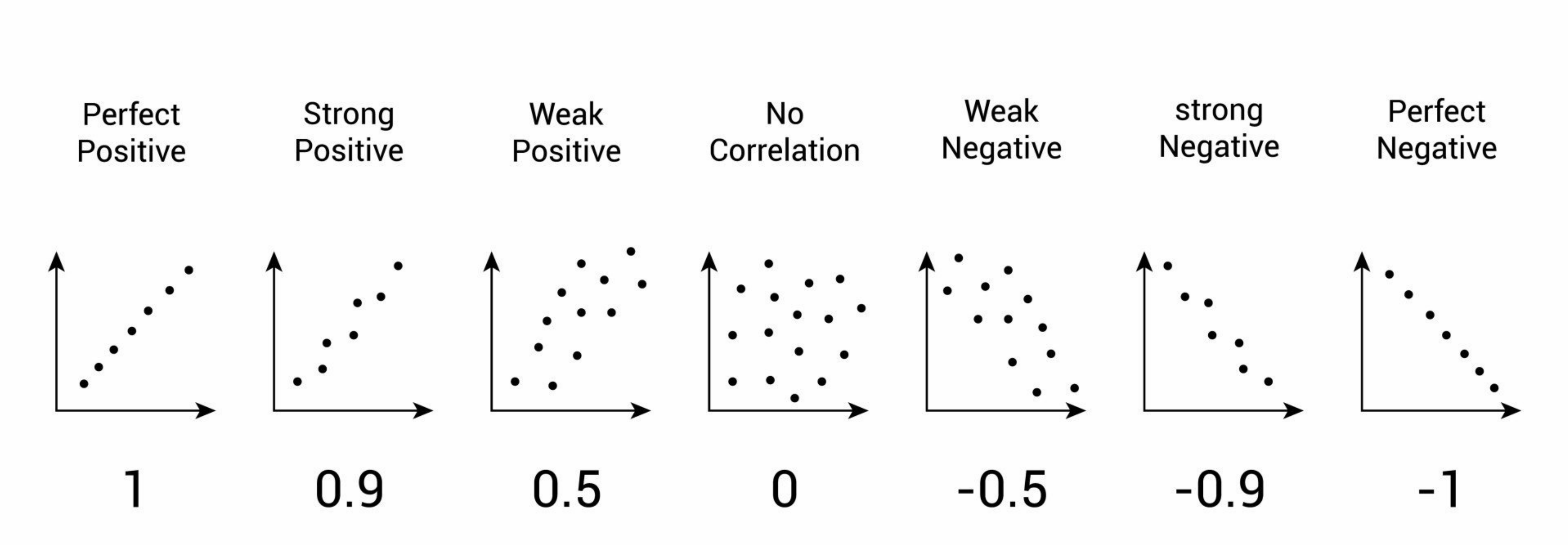
A scatter plot indicates the strength and direction of the correlation between the co-variables.



AI VIET NAM
@aivietnam.edu.vn

Correlation

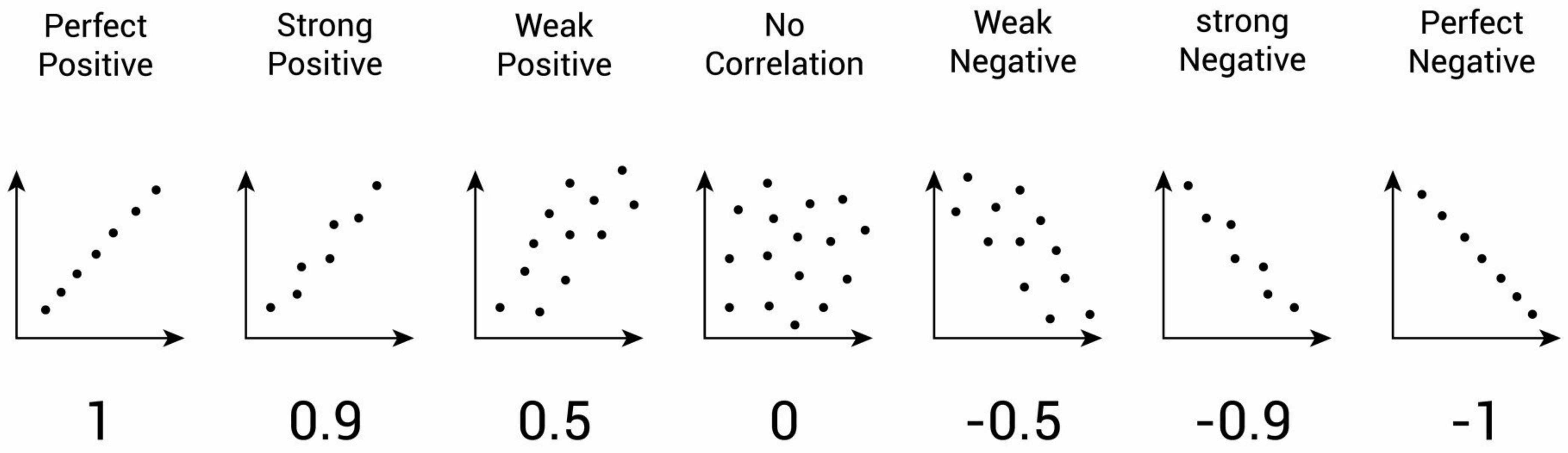
Instead of drawing a scatter plot, a correlation can be expressed numerically as a coefficient, ranging from -1 to +1. When working with continuous variables, the correlation coefficient to use is Pearson's r.



A correlation coefficient, often expressed as *r*, indicates a measure of the direction and strength of a relationship between two variables. When the *r* value is closer to +1 or -1, it indicates that there is a stronger linear relationship between the two variables.[1]

AI VIET NAM
@aivietnam.edu.vn

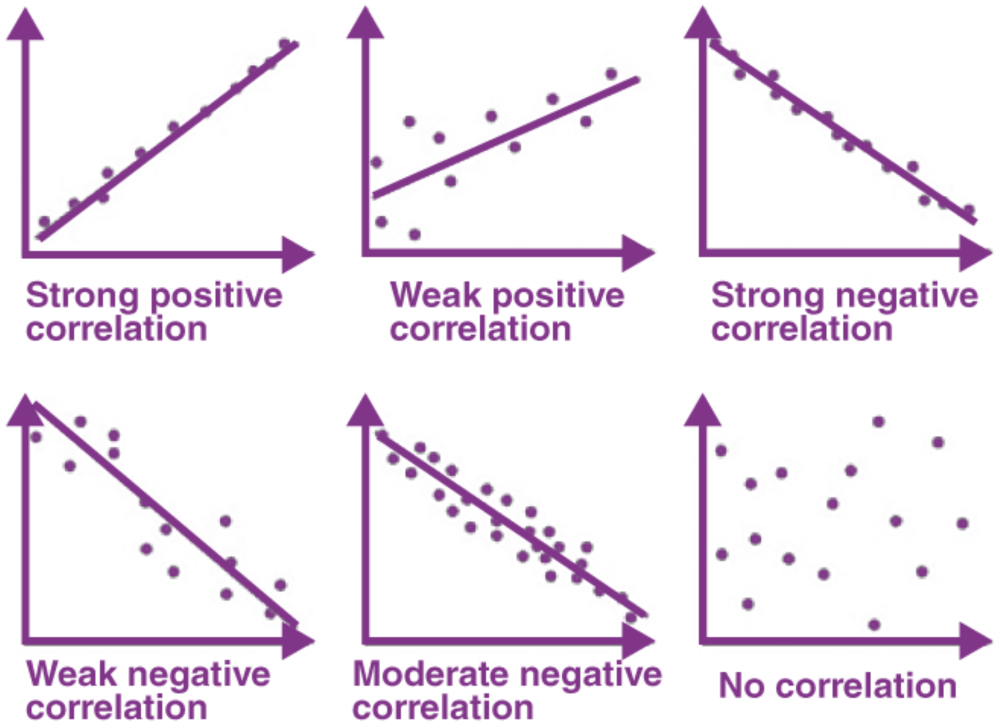
Correlation



A correlation of –1 indicates a perfect negative correlation, meaning that as one variable goes up, the other goes down. A correlation of +1 indicates a perfect positive correlation, meaning that as one variable goes up, the other goes up.

AI VIET NAM
@aivietnam.edu.vn

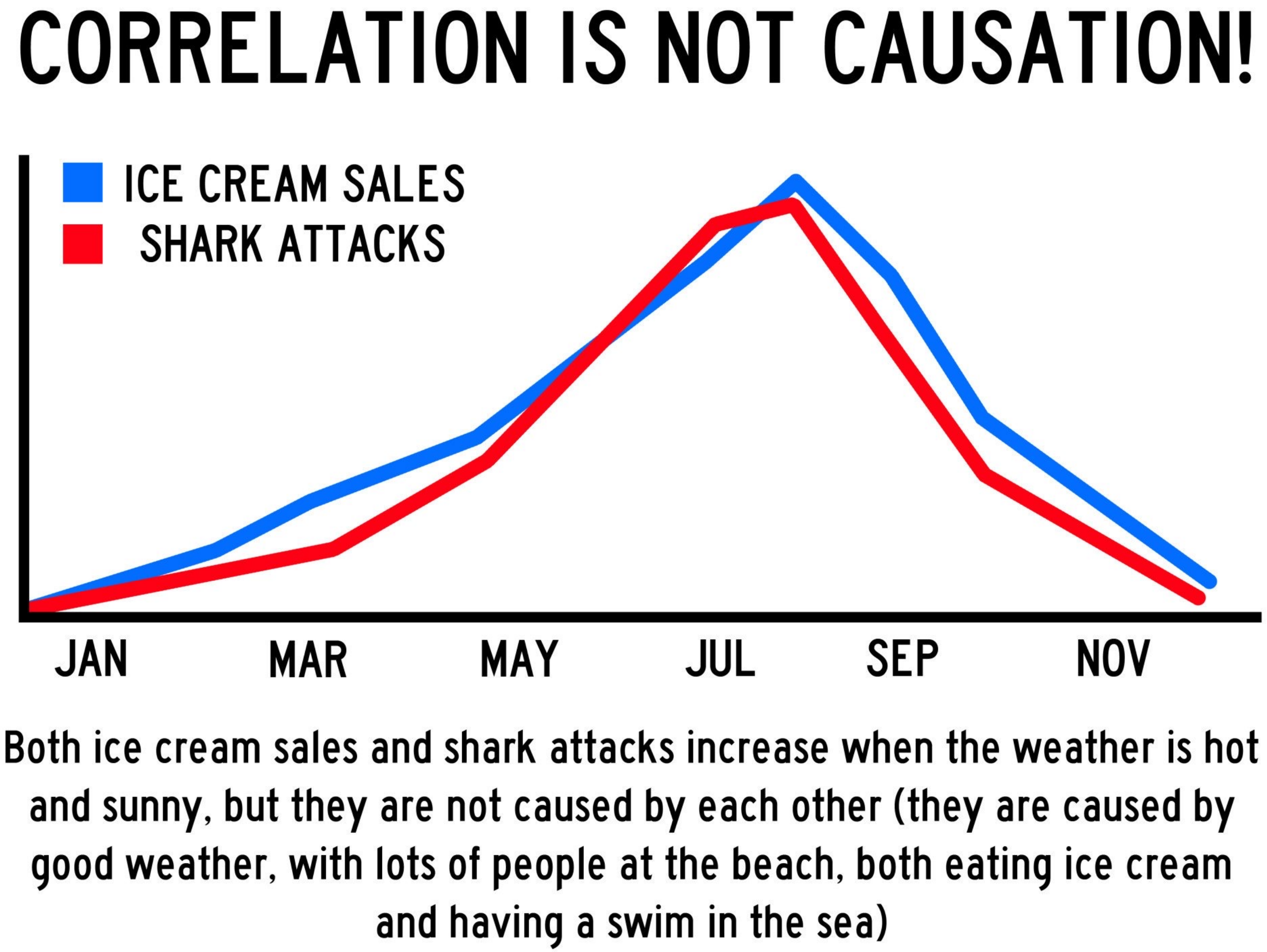
Correlation



A correlation of –1 indicates a perfect negative correlation, meaning that as one variable goes up, the other goes down. A correlation of +1 indicates a perfect positive correlation, meaning that as one variable goes up, the other goes up.

AI VIET NAM
@aivietnam.edu.vn

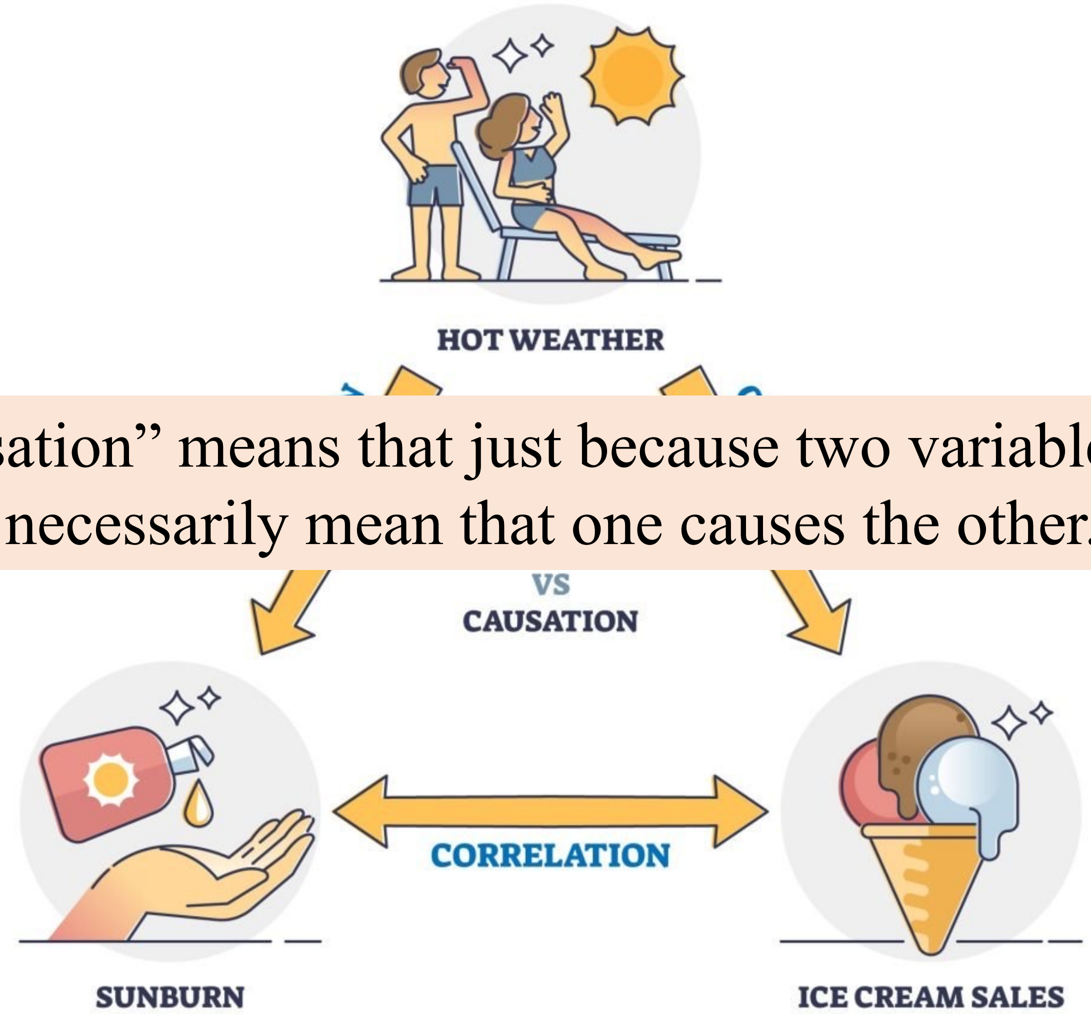
Correlation vs Causation



https://www.simplypsychology.org/correlation.html

AI VIET NAM
@aivietnam.edu.vn

Correlation vs Causation



"Correlation is not causation" means that just because two variables are related it does not necessarily mean that one causes the other.

https://www.simplypsychology.org/correlation.html

AI VIET NAM
@aivietnam.edu.vn

Correlation: Strengths

Correlation allows the researcher to investigate naturally occurring variables that may be unethical or impractical to test experimentally.

For example, it would be unethical to conduct an experiment on whether smoking causes lung cancer.

Correlation allows the researcher to clearly and easily see if there is a relationship between variables. This can then be displayed in a graphical form.

if there is a very strong association between two variables, Can we assume that they causes the other.

https://www.simplypsychology.org/correlation.html

Outline

- Mean, Variance and Standard Deviation: Review
- What is Correlation?
- What is Covariance?
- What is Correlation Coefficient
- Template Matching
- Summary

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

Ha Noi

Ho Chi Minh

Can Tho

Da Nang

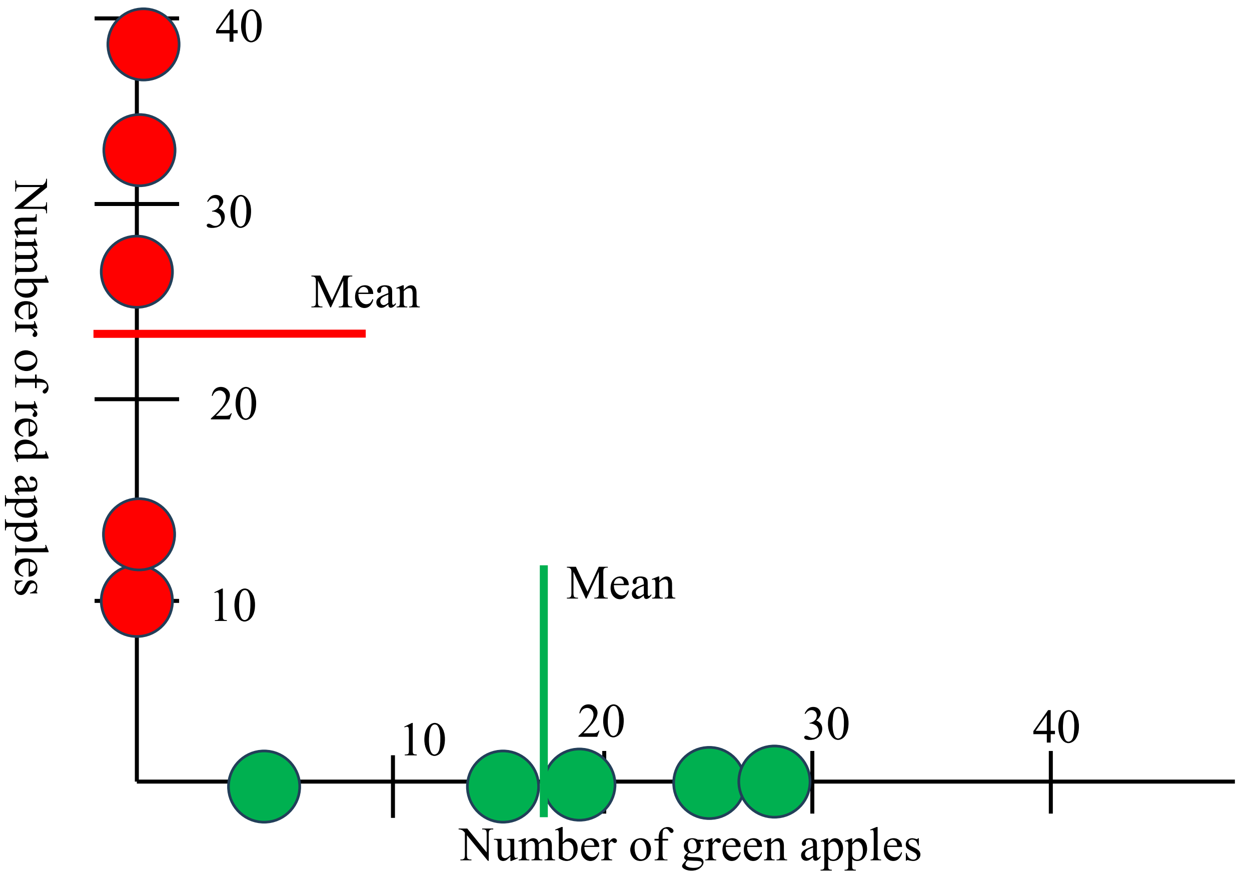
Binh Dinh

Count numbers of green apples and red apples in different supper markets

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2

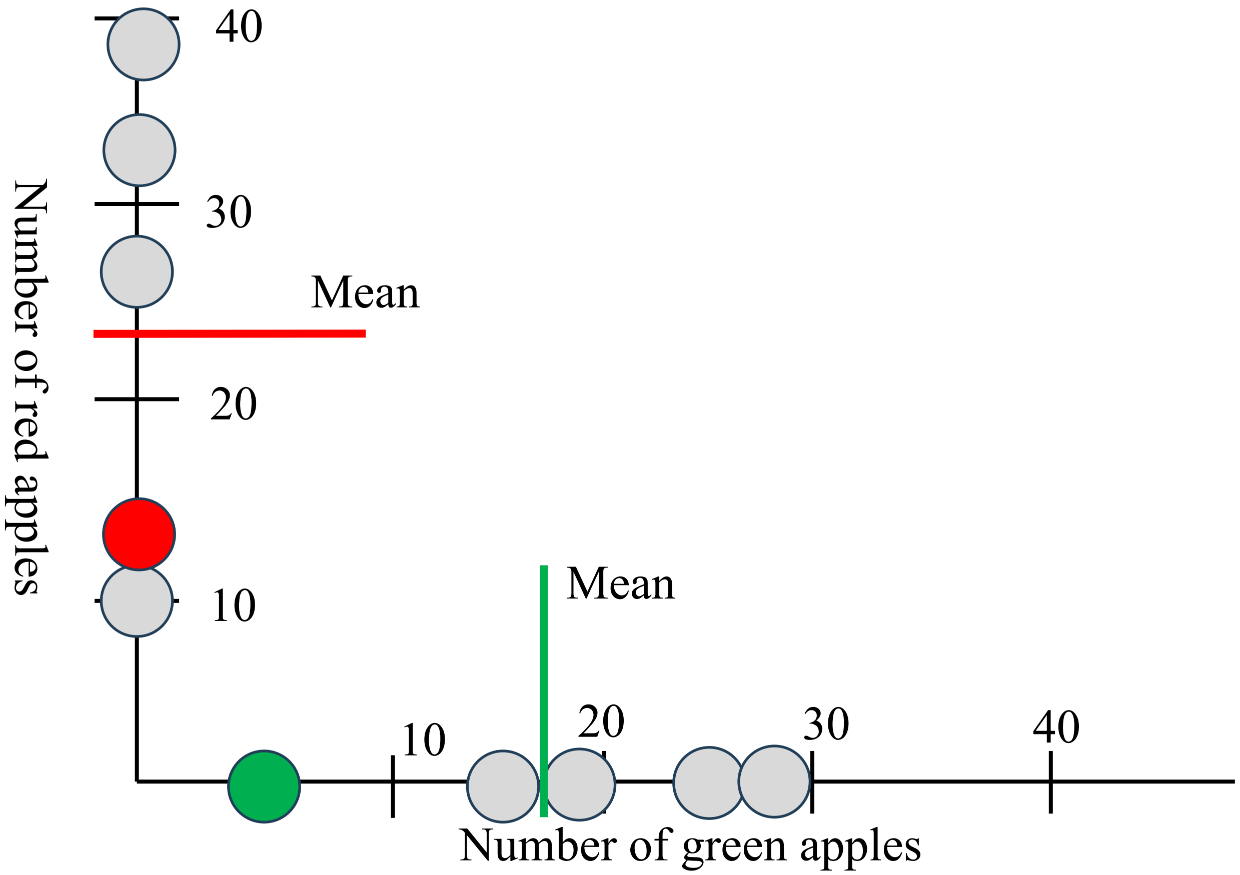


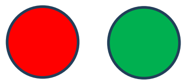
Count numbers of green apples and red apples in different supper markets

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



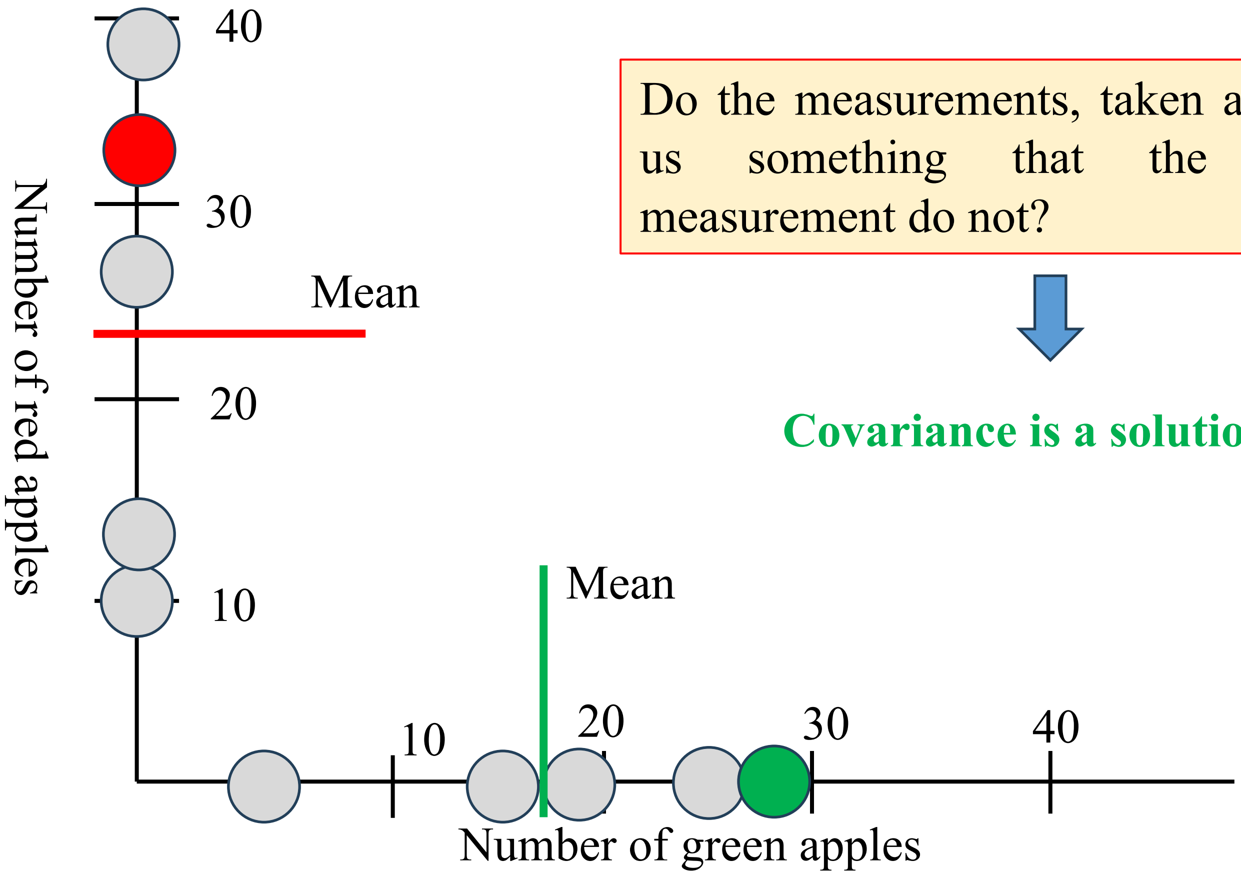


Same supermarket: Both measurements are less than their mean values.

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



Do the measurements, taken as pairs, tell us something that the individual measurement do not?

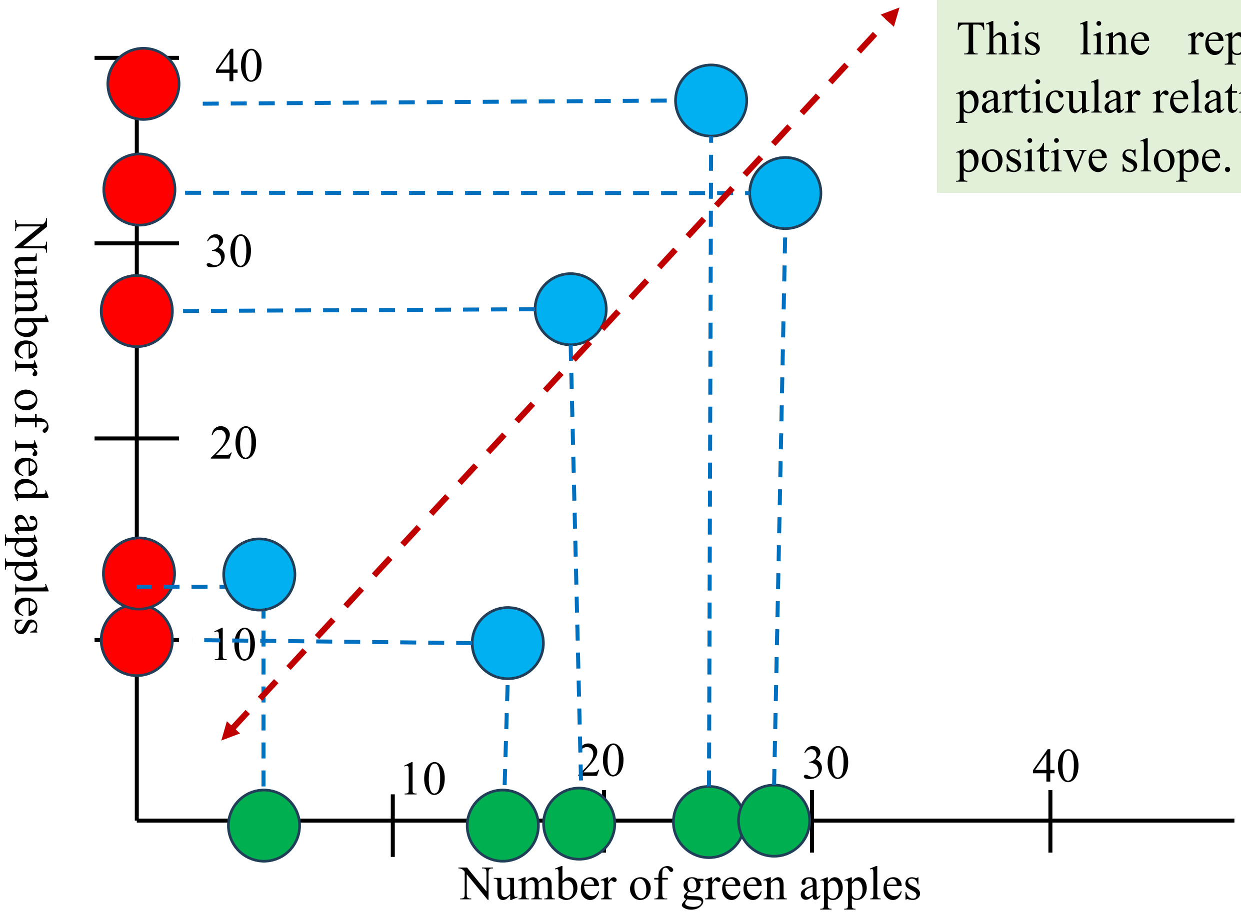
Covariance is a solution

Same supermarket: Both measurements are greater than their mean values.

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

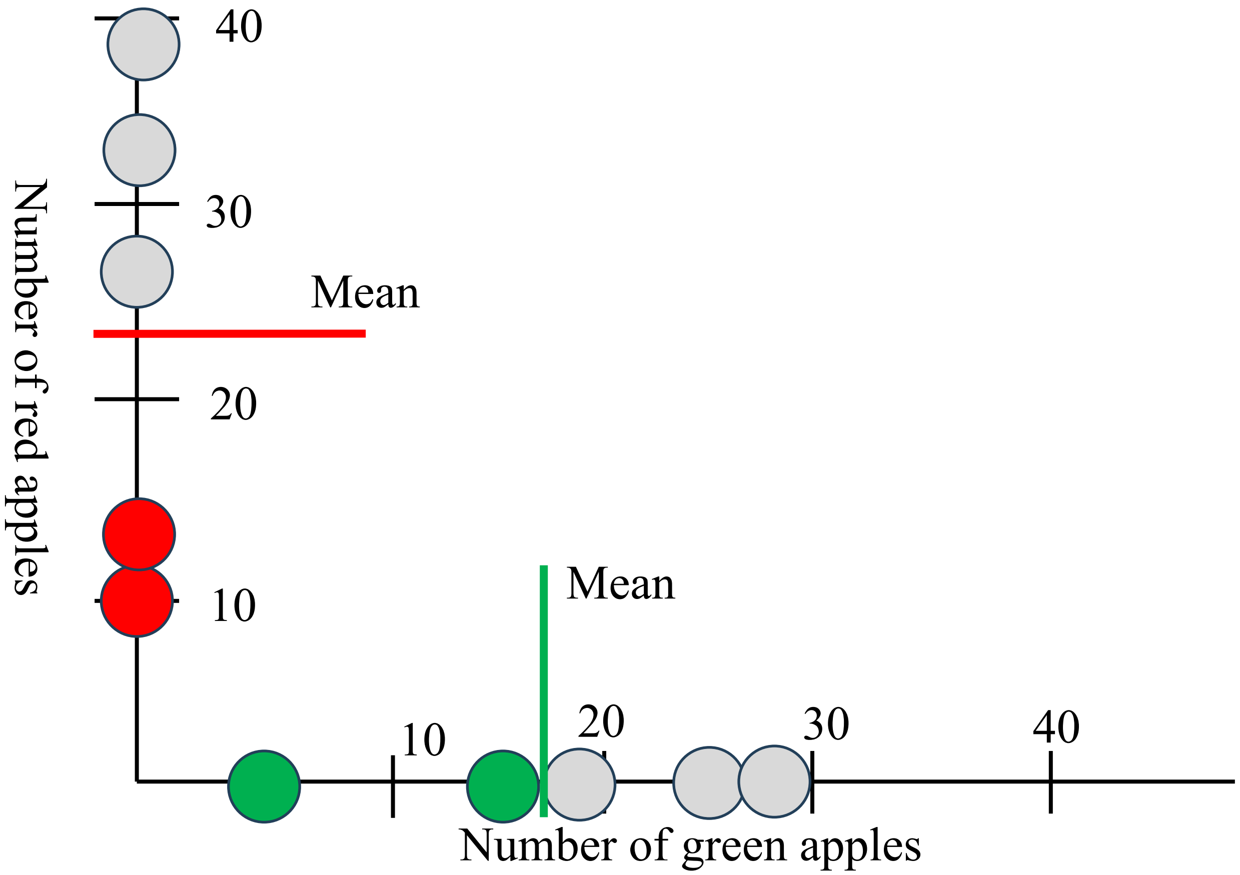
	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



This line represents this particular relationship has a positive slope.

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2

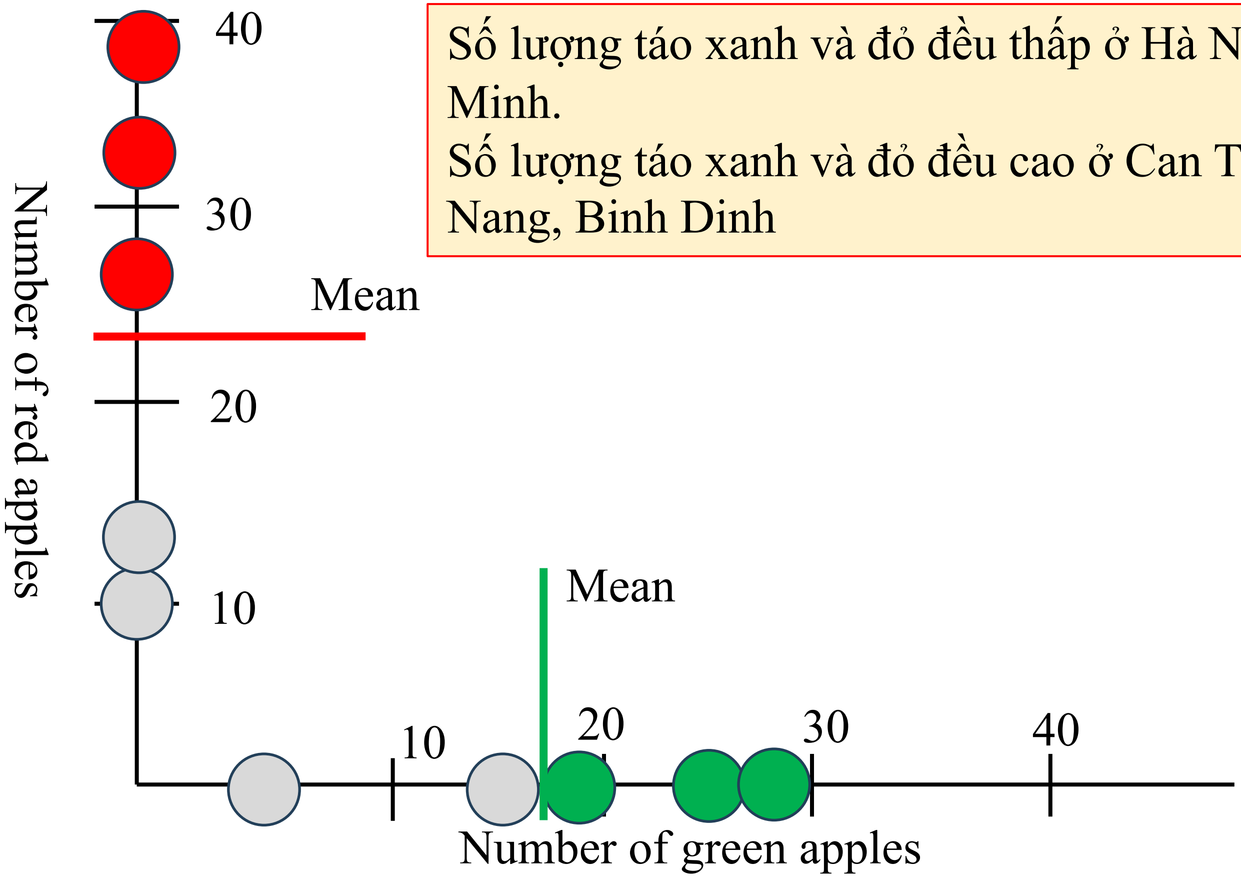


Two supermarkets, low amount of green apples => low amount of red apples

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



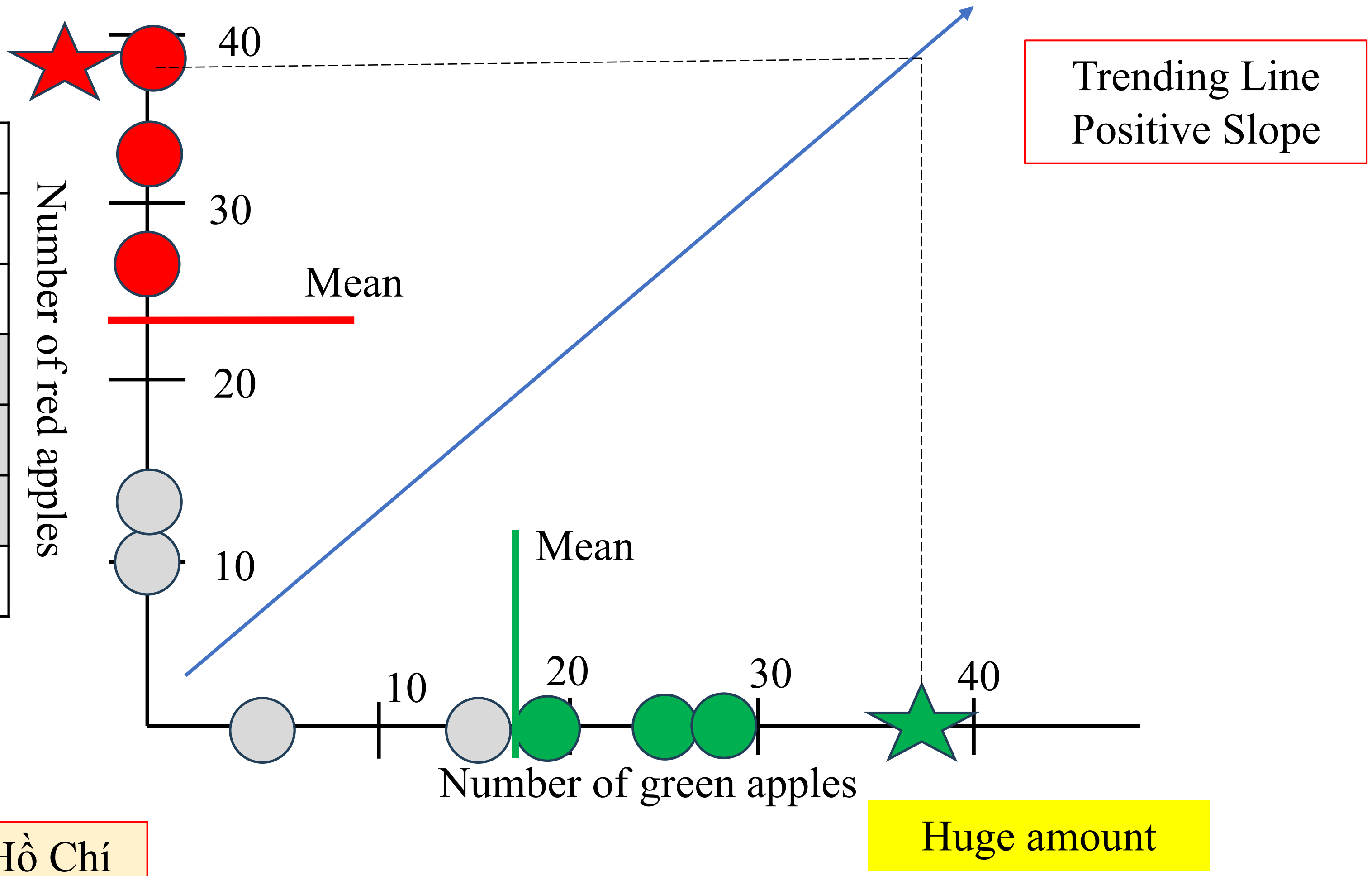
Số lượng táo xanh và đỏ đều thấp ở Hà Nội, Hồ Chí Minh.
Số lượng táo xanh và đỏ đều cao ở Can Tho, Da Nang, Binh Dinh

Three supermarkets, high amount of green apples => high amount of red apples

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

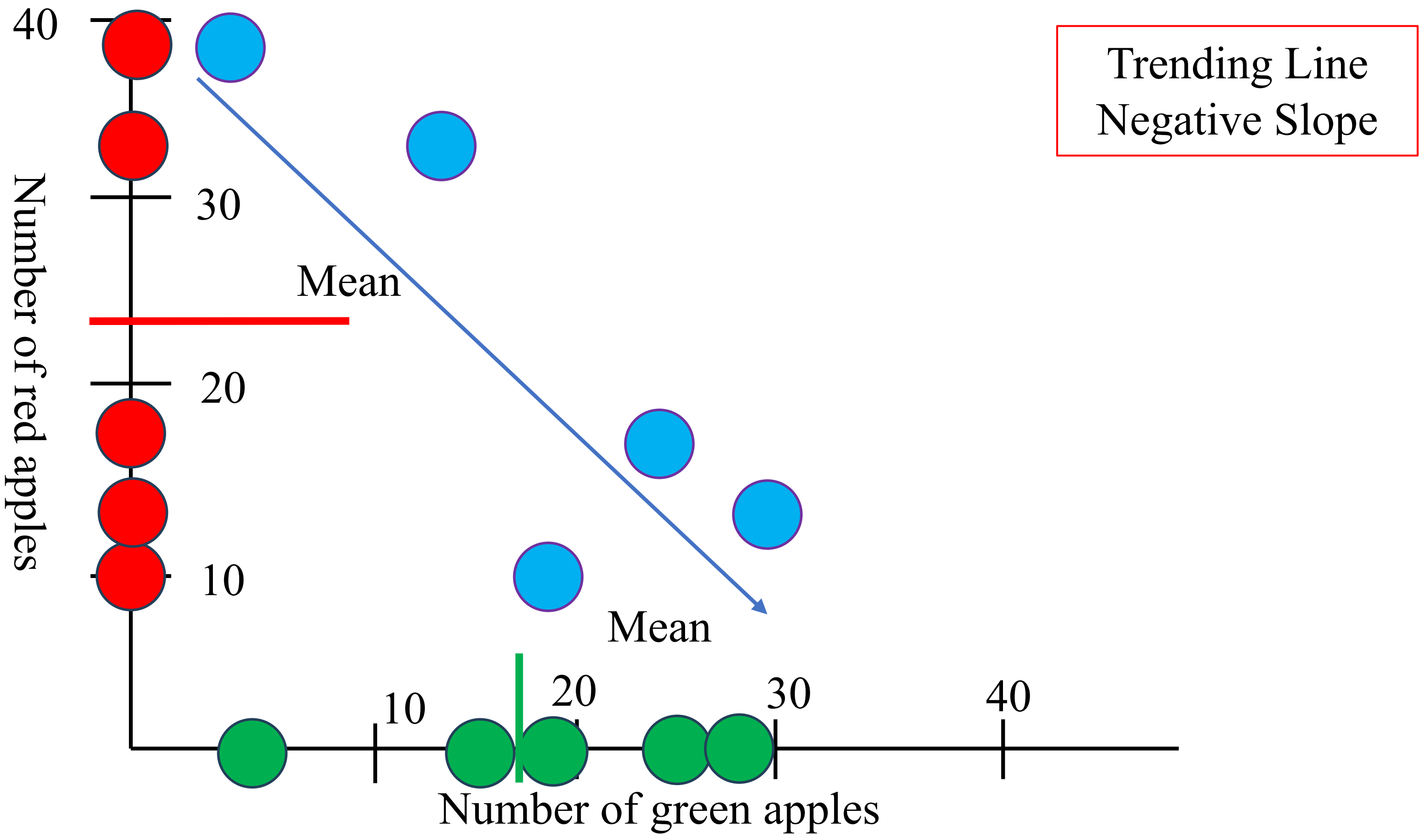
	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



Số lượng táo xanh và đỏ đều thấp ở Hà Nội, Hồ Chí Minh.
Số lượng táo xanh và đỏ đều cao ở Can Tho, Da Nang, Binh Dinh

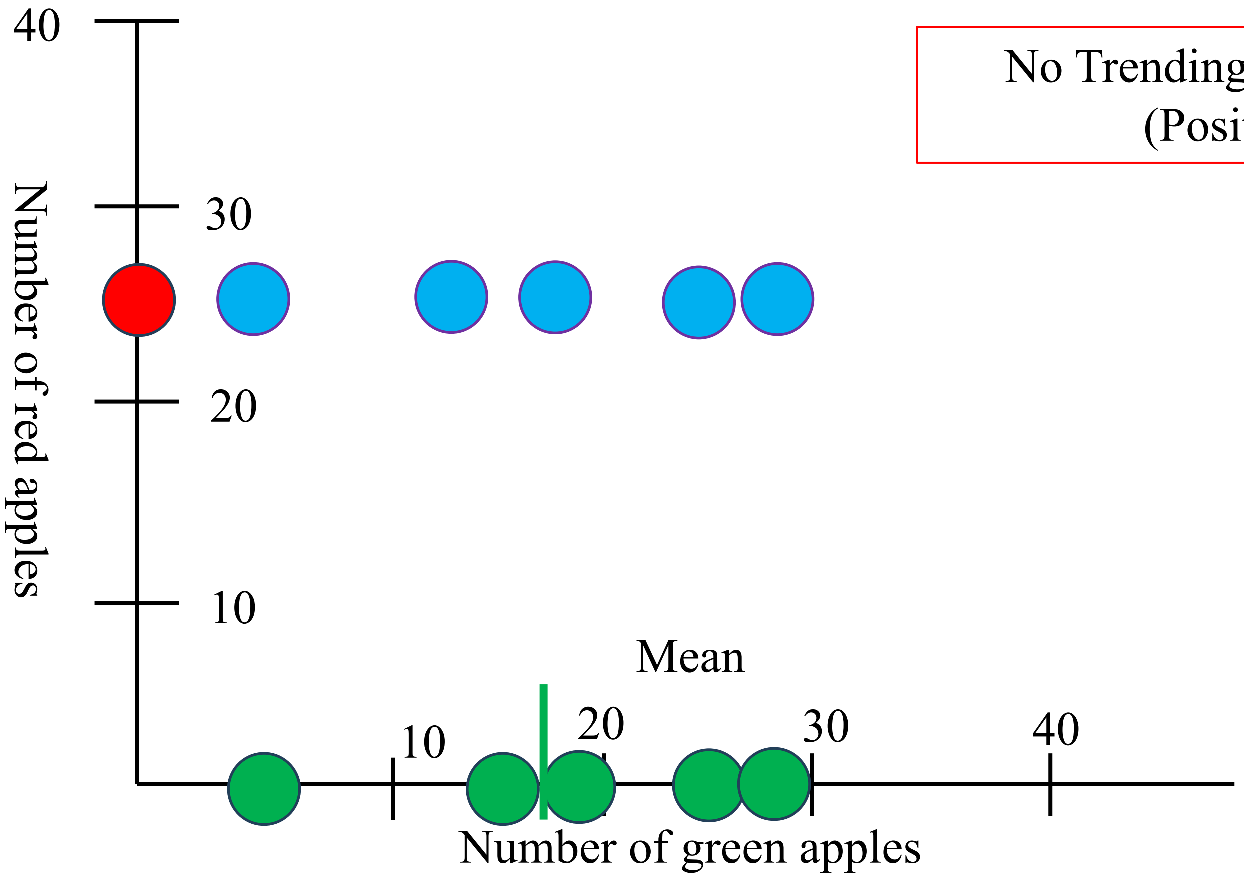
AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

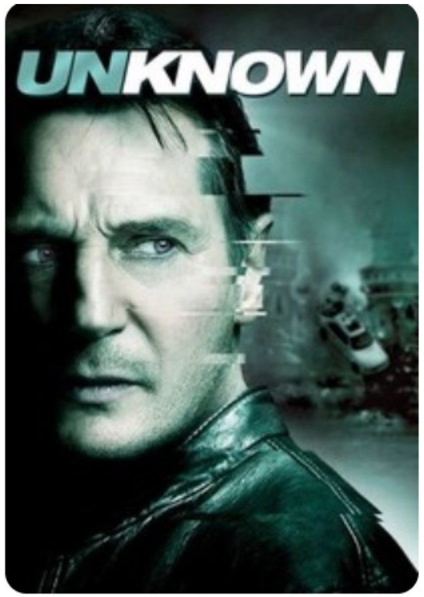


AI VIET NAM
@aivietnam.edu.vn

What is Covariance?



No Trending Line (No Relationship)
(Positive or Negative)

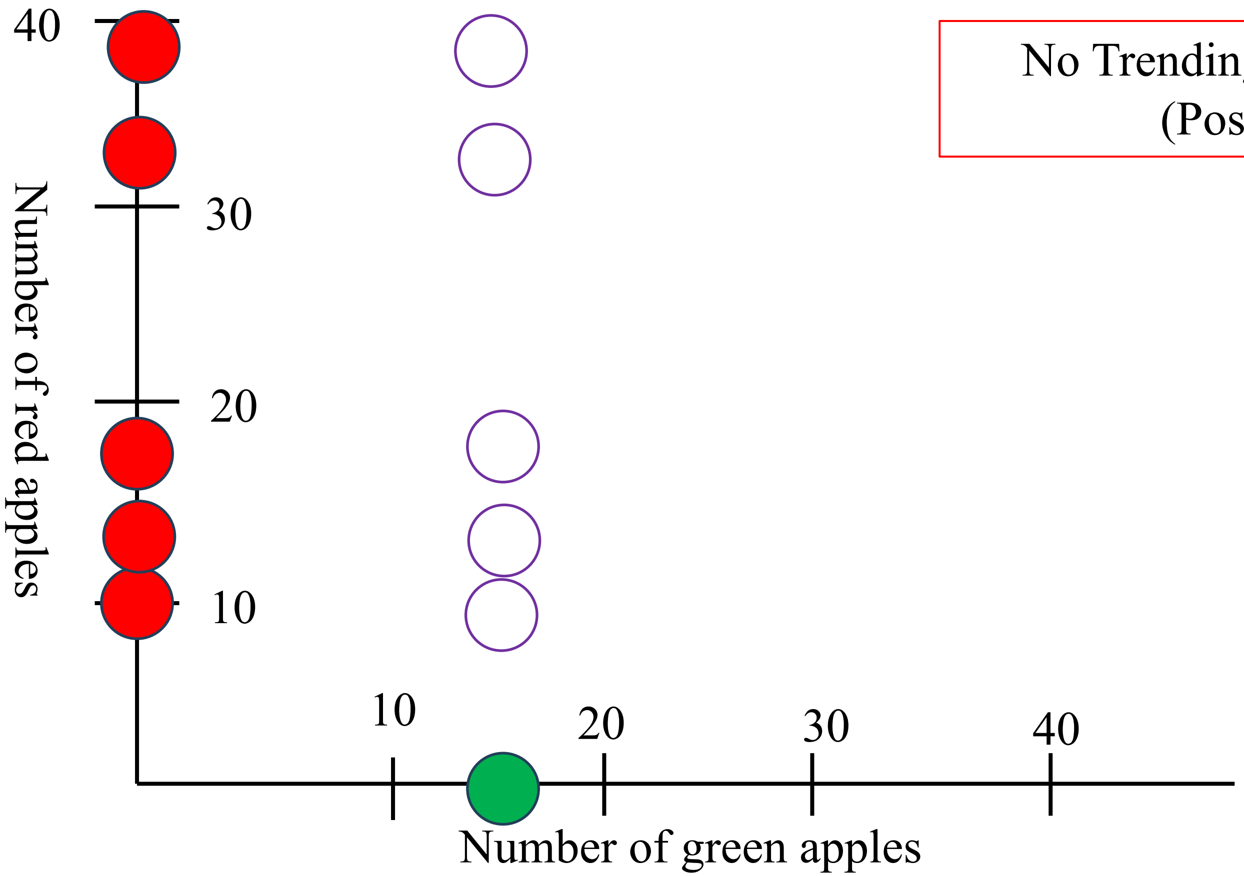


Cho trước số lượng táo đỏ nhiều ở một suppermarket, số lượng táo xanh ở cùng suppermarket như thế nào?

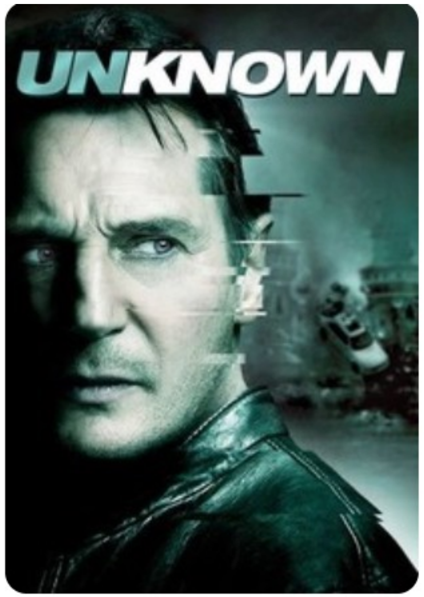
AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

Ý tưởng chính của Covariance là làm sao nhận biết được 3 mối liên hệ này giữa Táo Xanh và Táo Đỏ



No Trending Line (No Relationship)
(Positive or Negative)



Cho trước số lượng táo xanh nhiều ở một suppermarket, số lượng táo đỏ ở cùng suppermarket như thế nào?

AI VIET NAM
@aivietnam.edu.vn

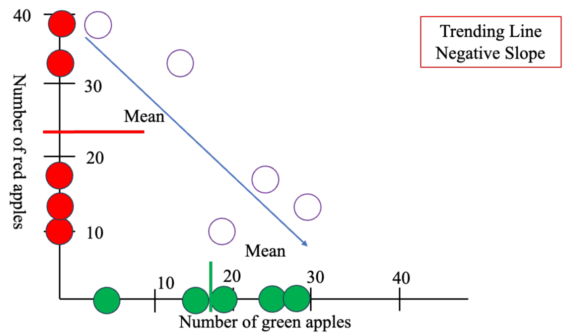
What is Covariance?

Ý tưởng chính của Covariance là làm sao nhận biết được 3 mối liên hệ giữa Táo Xanh và Táo Đỏ

AI VIETNAM
All-In-One Course

What is Covariance?

Negative trend

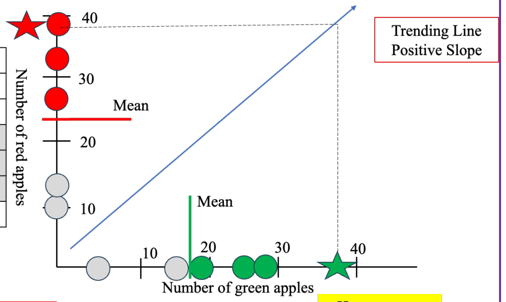


Positive trend

AI VIETNAM
All-In-One Course

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2

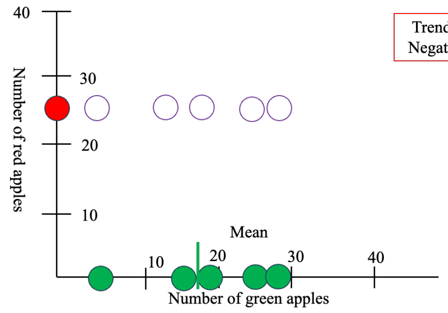


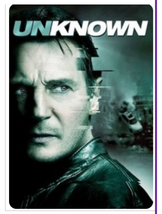
Số lượng táo xanh và đỏ đều thấp ở Hà Nội, Hồ Chí Minh.
Số lượng táo xanh và đỏ đều cao ở Can Tho, Da Nang, Binh Dinh

No trend

AI VIETNAM
All-In-One Course

What is Covariance?



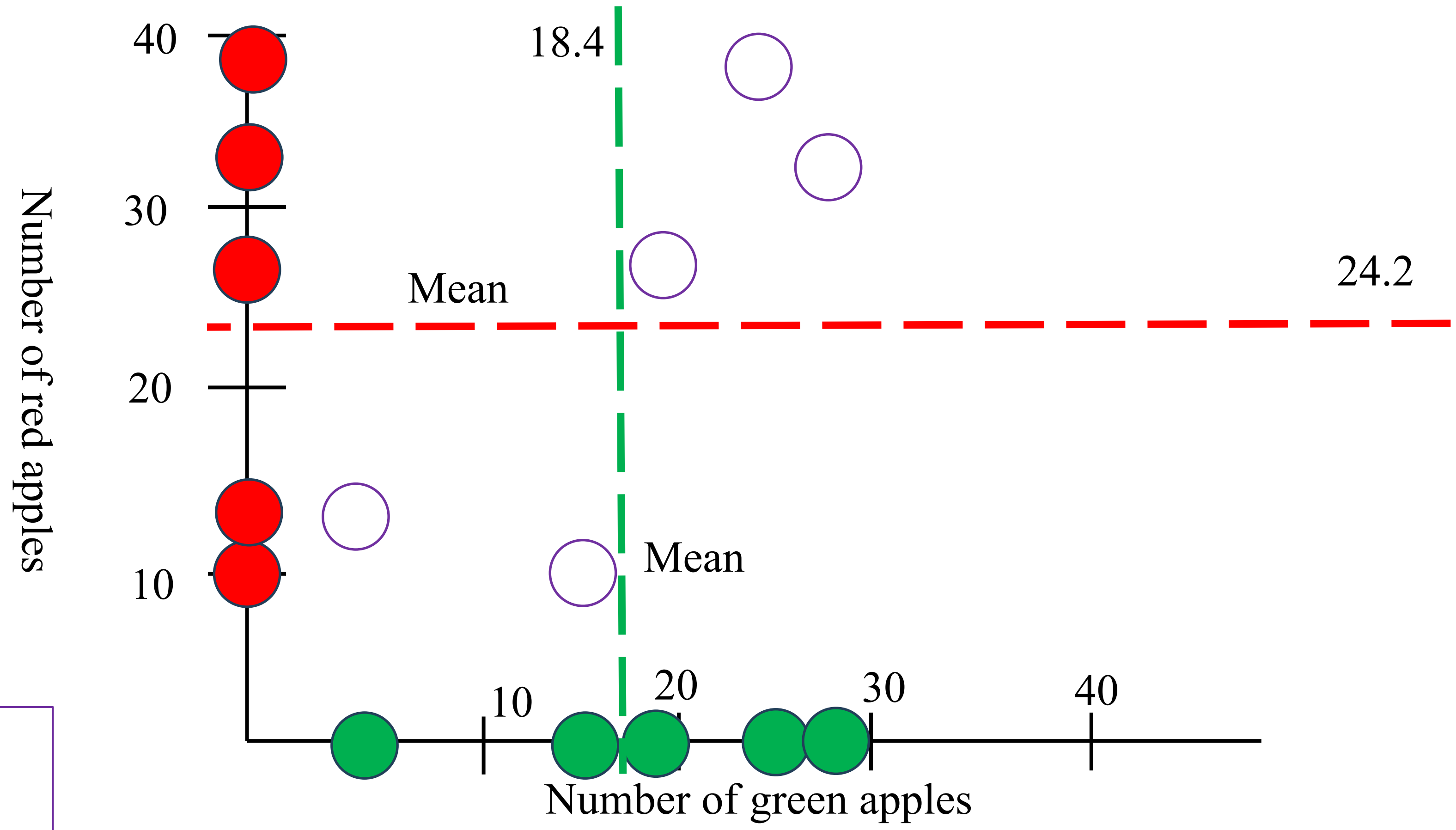


Cho trước số lượng táo đỏ nhiều ở một suppermarket, số lượng táo xanh ở cùng suppermarket như thế nào?

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



Covariance Formula

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$

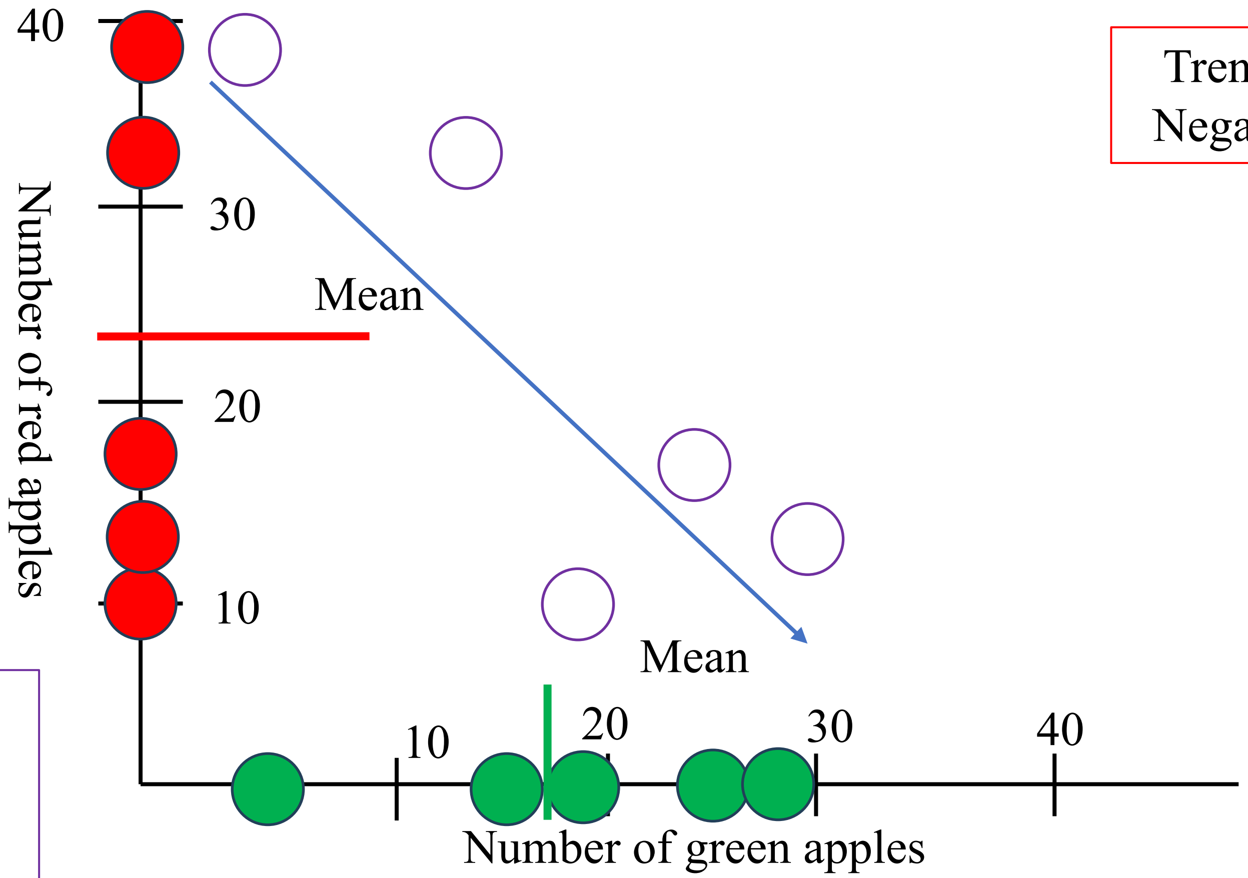
Cov(amounts green apple, amount red apple) > 0

Because Cov > 0, we classify as the positive trend

Why?

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?



Trending Line
Negative Slope

Cov(amounts green apple, amount red apple) < 0

Because Cov < 0, we classify as the negative trend

Covariance Formula

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

Trending Line
Negative Slope

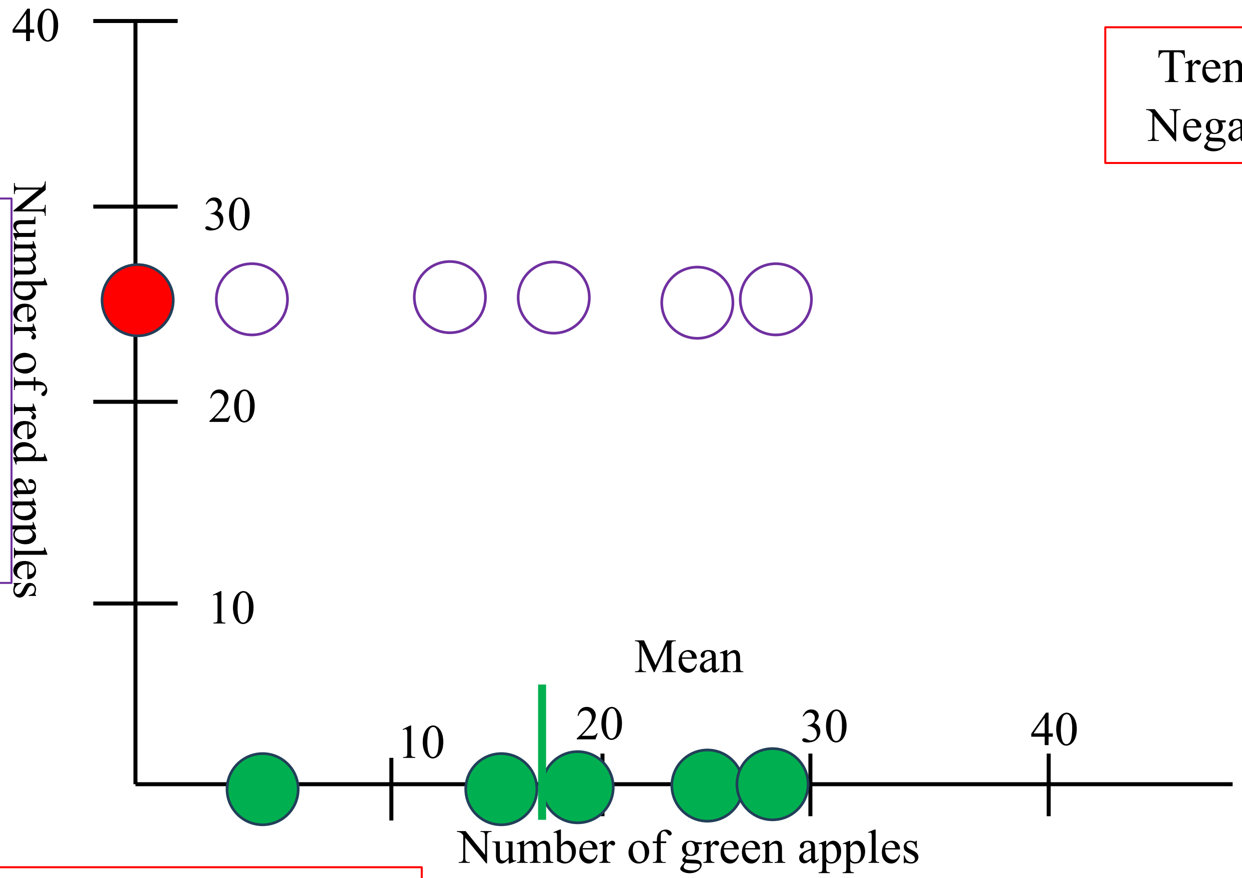
Covariance Formula

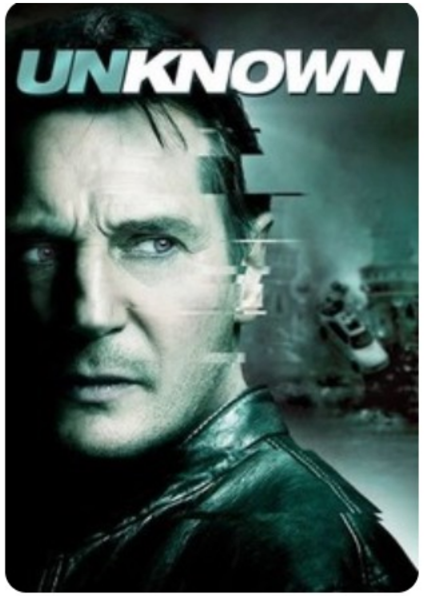
For Population

$$\text{Cov}(x,y) = \frac{\Sigma\,(x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$\text{Cov}(x,y) = \frac{\Sigma\,(x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$





Cov(amounts green apple, amount red apple) = 0

Because Cov = 0, we classify as the no trend

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

	Green apple	**Red apple**
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2

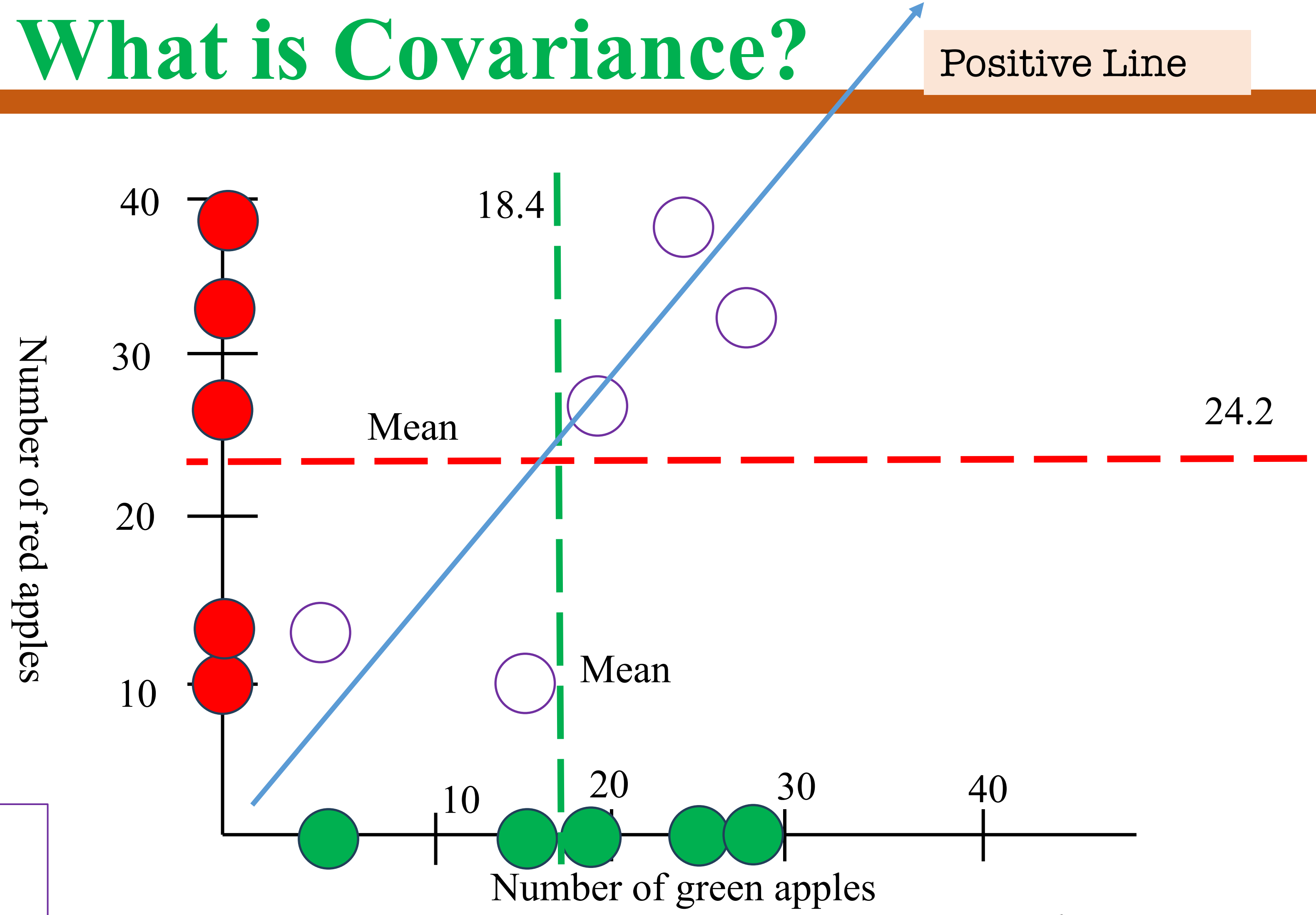
Covariance Formula

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$



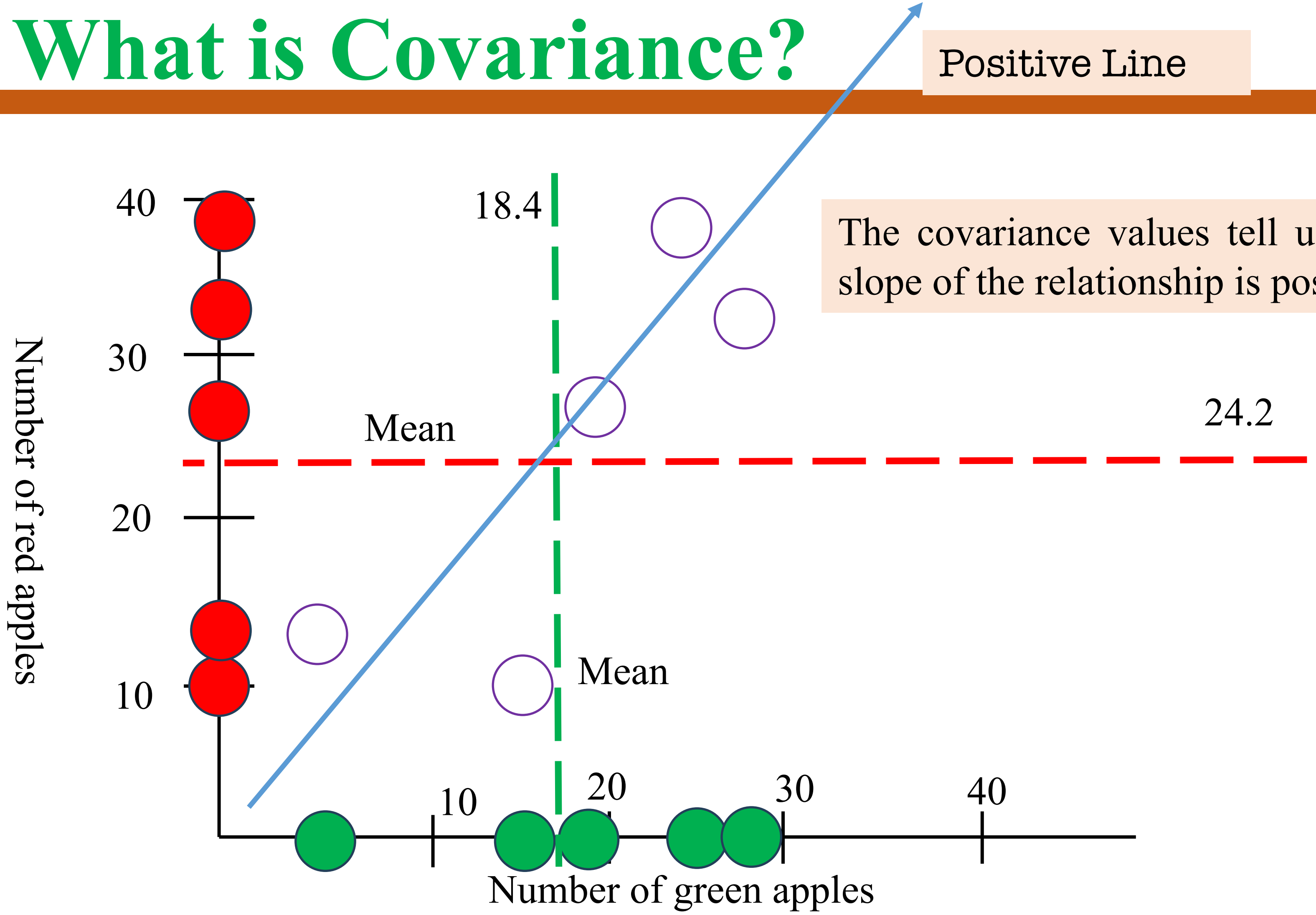
Why?

Covariance is difficult to interpret and depends on the context

AI VIET NAM
@aivietnam.edu.vn

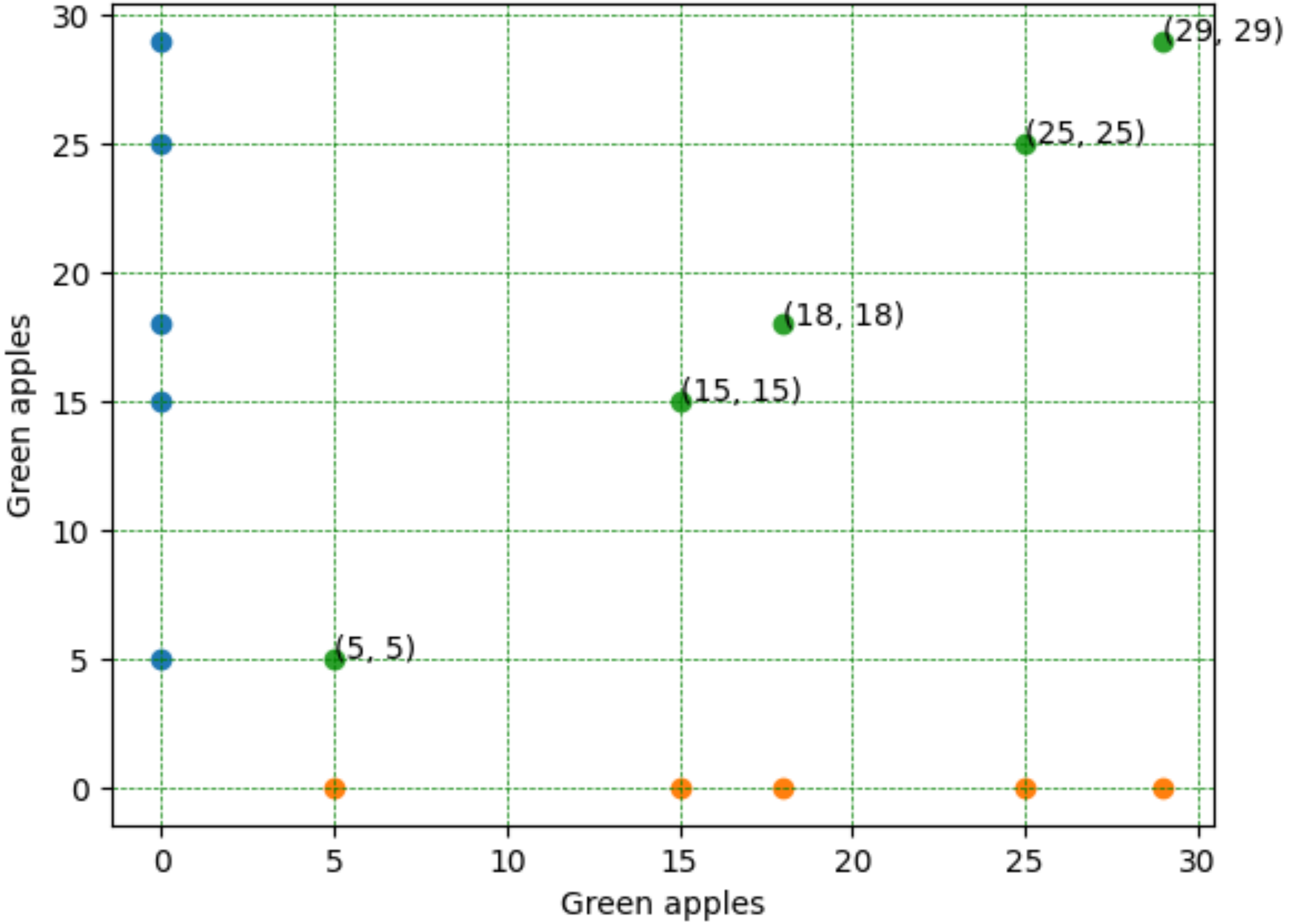
What is Covariance?

	Green apple	Red apple
Ha Noi	5	12
Ho Chi Minh	15	10
Can Tho	18	28
Da Nang	25	38
Binh Dinh	29	33
	18.4	24.2



The covariance values tell us the slope of the relationship is postive

The covariance value does not tell us if the slope of the line representing the relationship is steep or not steep.
The covariance values does not tell us if the points are relativel close the trending line or not.



Covariance value is senstive to scale data

```python
import numpy as np
y = [5, 15, 18, 25, 29]
x = [5, 15, 18, 25, 29]
covariance = np.cov(x, y)[0][1]
print(np.round(covariance,2))
```

```
86.8
```

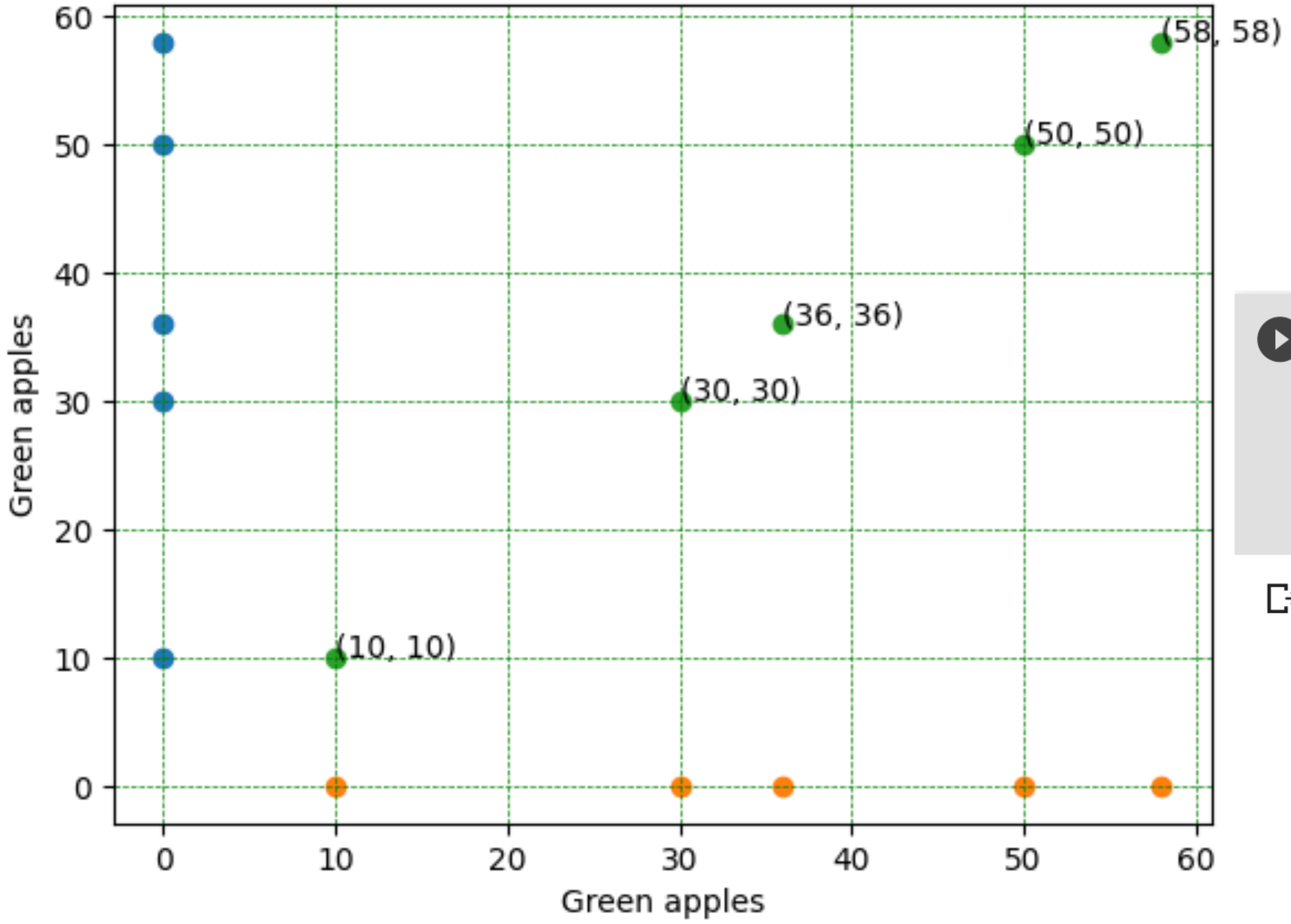
Covariance Formula

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N-1)}$$




```python
import numpy as np
y = [5*2, 15*2, 18*2, 25*2, 29*2]
x = [5*2, 15*2, 18*2, 25*2, 29*2]
covariance = np.cov(x, y)[0][1]
print(np.round(covariance,2))
```

```
347.2
```

Covariance Formula

For Population

$$Cov(x,y) = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$

AI VIET NAM
@aivietnam.edu.vn

What is Covariance?

Why?

The covariance is hard to interpret

AI VIET NAM
@aivietnam.edu.vn

Covariance → Correlation Coefficient

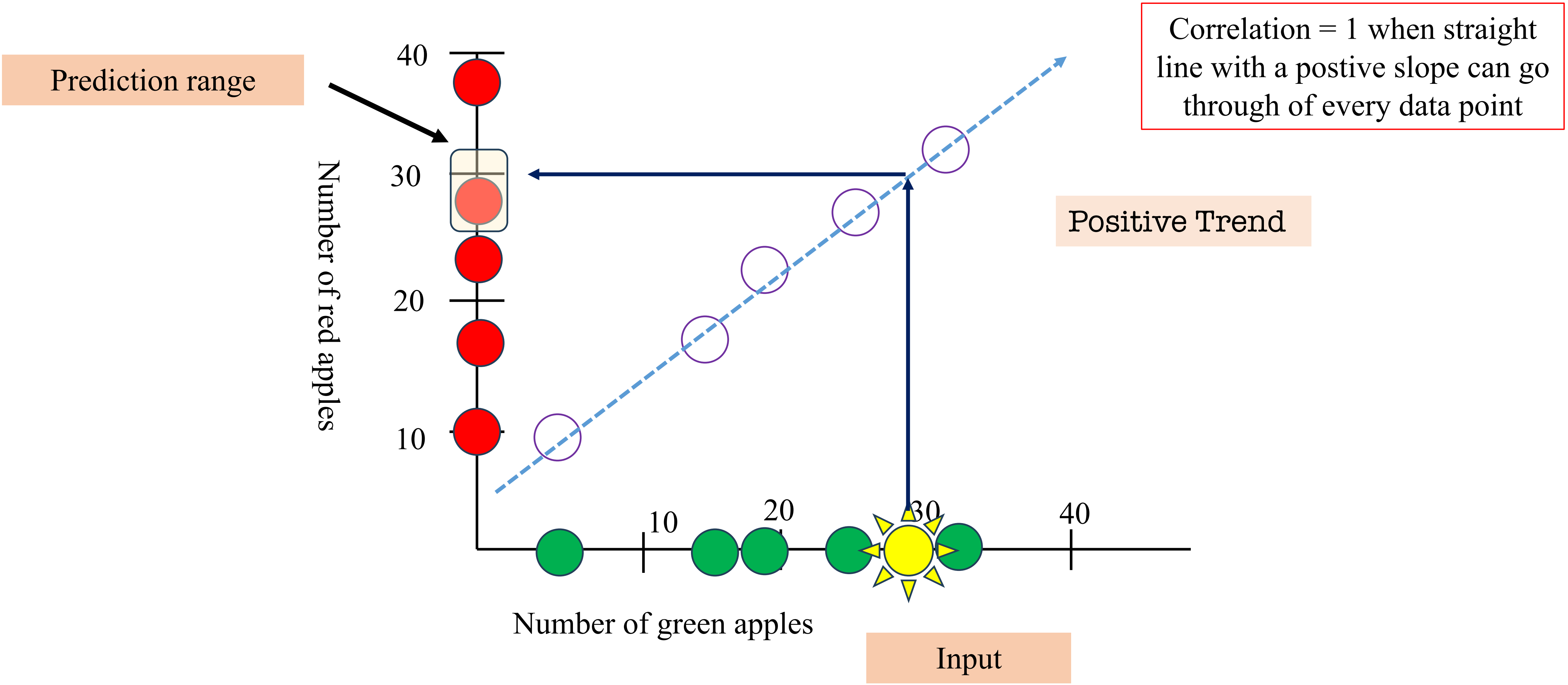
The covariance is hard to interpret, but it is computational stepping stones (Correlation Coefficient, or Princial Component Analysis, ...)

Correlation describes relationships: how strong is the correlation [-1,1]

Correlation is not sensitive to the scale of the data

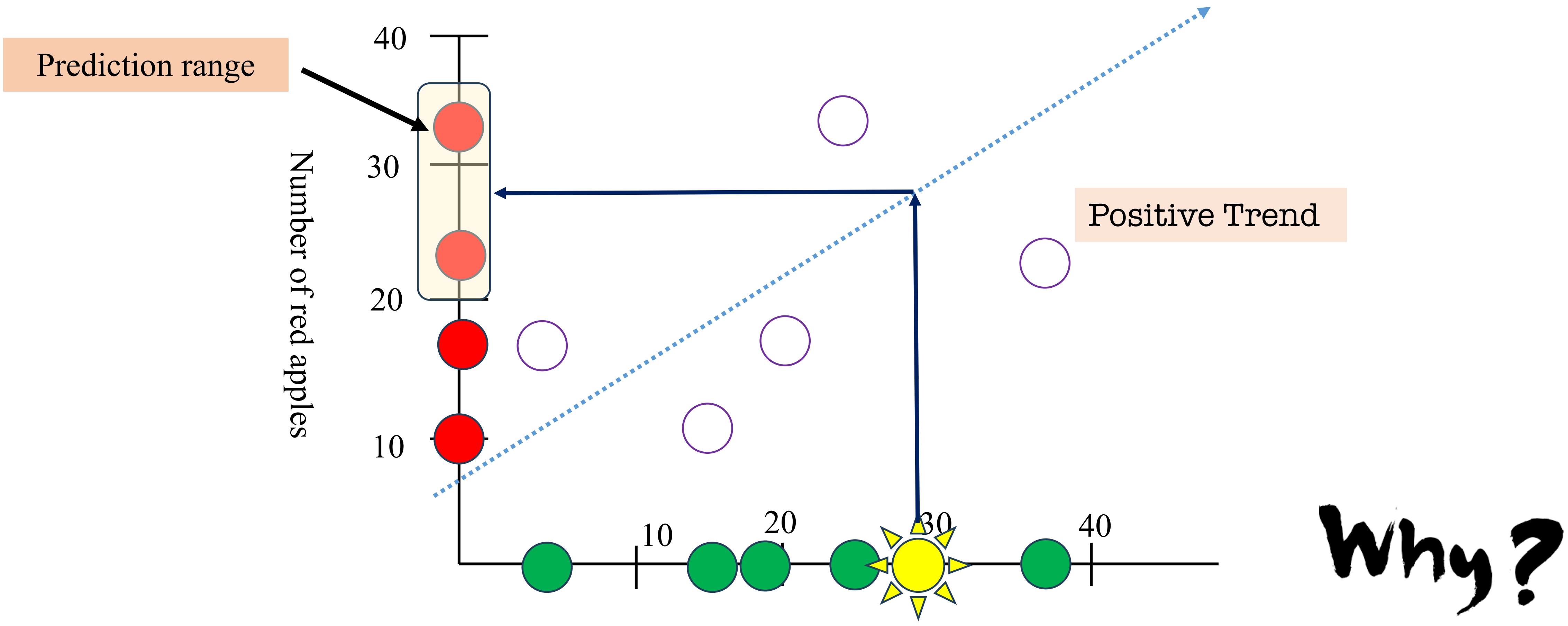
AI VIET NAM
@aivietnam.edu.vn

Stronger Relationship



AI VIET NAM
@aivietnam.edu.vn

Weaker Relationship



Correlation does not depend on the scale of the data

Outline

- ➢ **Mean, Variance and Standard Deviation: Review**
- ➢ **What is Correlation?**
- ➢ **What is Covariance?**
- ➢ **What is Correlation Coefficient**
- ➢ **Template Matching**
- ➢ **Summary**

AI VIET NAM
@aivietnam.edu.vn

Common Correlation coefficients

Correlation coefficient	Type of relationship	Levels of measurement	Data distribution
Pearson's r	Linear	Two quantitative (interval or ratio) variables	Normal distribution
Spearman's rho	Non-linear	Two ordinal, interval or ratio variables	Any distribution
Point-biserial	Linear	One dichotomous (binary) variable and one quantitative (interval or ratio) variable	Normal distribution
Cramér's V (Cramér's φ)	Non-linear	Two nominal variables	Any distribution
Kendall's tau	Non-linear	Two ordinal, interval or ratio variables	Any distribution

AI VIET NAM
@aivietnam.edu.vn

Common Correlation coefficients

Correlation coefficient	Type of relationship	Levels of measurement	Data distribution
Pearson's r	Linear	Two quantitative (interval or ratio) variables	Normal distribution
Spearman's rho	Non-linear	Two ordinal, interval or ratio variables	Any distribution
Point-biserial	Linear	One dichotomous (binary) variable and one quantitative (interval or ratio) variable	Normal distribution
Cramér's V (Cramér's φ)	Non-linear	Two nominal variables	Any distribution
Kendall's tau	Non-linear	Two ordinal, interval or ratio variables	Any distribution

Correlation Coefficient

$$\rho_{X,Y} = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y}$$

Pearson's r

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

❖ Definition

Công thức: Gọi x,y là hai biến ngẫu nhiên

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

Tính chất 1

$$-1 \le \rho_{xy} \le 1$$

Tương quan nghịch ← → Tương quan thuận

Tính chất 2

$$\rho_{xy} = \rho_{uv}$$

trong đó

$$u = ax + b$$
$$v = cy + d$$

Ví dụ 1

x = [7, 18, 29, 2, 10, 9, 9]
y = [1, 6, 12, 8, 6, 21, 10]

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n*818 - 84*64}{\sqrt{n*1480 - 7056}\sqrt{n*822 - 4096}} = 0.149$$

Ví dụ 2

u=2*x-14 = [0, 22, 44, -10, 6, 4, 4]
v=y+2 = [3, 8, 14, 10, 8, 23, 12]

$$\rho_{uv} = \frac{E[(u-\mu_u)(v-\mu_v)]}{\sqrt{var(u)}\sqrt{var(v)}}$$

$$= \frac{n*880 - 70*78}{\sqrt{n*2588 - 4900}\sqrt{n*1106 - 6084}} = 0.149$$

AI VIET NAM
@aivietnam.edu.vn

Correlation Coefficient

❖ Properties

Công thức: Gọi x,y là hai biến ngẫu nhiên

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

Tính chất: Given a, b > 0; we have

$$\rho_{xy}(x,y) = \rho_{xy}(ax,by)$$

$$\rho_{xy}(ax,by) = \frac{E[(ax-\mu_{ax})(by-\mu_{by})]}{\sqrt{var(ax)}\sqrt{var(by)}}$$

$$= \frac{n(\sum_i ax_i by_i) - (\sum_i ax_i)(\sum_i by_i)}{\sqrt{n\sum_i a^2 x_i^2 - (\sum_i ax_i)^2}\sqrt{n\sum_i b^2 y_i^2 - (\sum_i by_i)^2}}$$

$$= \frac{n(ab\sum_i x_i y_i) - (a\sum_i x_i)(b\sum_i y_i)}{\sqrt{na^2\sum_i x_i^2 - a^2(\sum_i x_i)^2}\sqrt{nb^2\sum_i y_i^2 - b^2(\sum_i y_i)^2}}$$

$$= \frac{ab\left[n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)\right]}{a\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\; b\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}} = \rho_{xy}$$

AI VIET NAM
@aivietnam.edu.vn

Correlation Coefficient

❖ Properties

Công thức: Gọi x,y là hai biến ngẫu nhiên

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

Tính chất: Given c and d are two constants, we have

$$\rho_{xy}(x,y) = \rho_{xy}(x+c, y+d)$$

$$\rho_{xy}(x+c, y+d) = \frac{E[((x+c)-\mu_{(x+c)})((y+d)-\mu_{(y+d)})]}{\sqrt{var(x+c)}\sqrt{var(y+d)}}$$

$$= \frac{E[((x+c)-(\mu_x+c))((y+d)-(\mu_y+d))]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}} = \rho_{xy}$$

AI VIET NAM
@aivietnam.edu.vn

Correlation Coefficient

Công thức: Gọi x,y là hai biến ngẫu nhiên

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

```
def find_corr_x_y(x,y):                                   #1
    n = len(x)                                            #2
    prod = []
    for xi,yi in zip(x,y):                                #3
        prod.append(xi*yi)

    sum_prod_x_y = sum(prod)                              #4

    sum_x = sum(x)
    sum_y = sum(y)

    squared_sum_x = sum_x**2
    squared_sum_y = sum_y**2

    x_square = []
    for xi in x:
        x_square.append(xi**2)
    x_square_sum = sum(x_square)

    y_square=[]
    for yi in y:
        y_square.append(yi**2)
    y_square_sum = sum(y_square)

    # Use formula to calculate correlation                #5
    numerator = n*sum_prod_x_y - sum_x*sum_y
    denominator_term1 = n*x_square_sum - squared_sum_x
    denominator_term2 = n*y_square_sum - squared_sum_y
    denominator = (denominator_term1*denominator_term2)**0.5
    correlation = numerator/denominator

    return correlation
```

Công thức: Gọi x,y là hai biến ngẫu nhiên

$$\rho_{xy} = \frac{E[(x-\mu_x)(y-\mu_y)]}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$= \frac{n(\sum_i x_i y_i) - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n\sum_i x_i^2 - (\sum_i x_i)^2}\sqrt{n\sum_i y_i^2 - (\sum_i y_i)^2}}$$

```python
import numpy as np

x = [7, 18, 29, 2, 10, 9, 9]
y = [1, 6, 12, 8, 6, 21, 10]

np.corrcoef(x, y)
```

```
array([[1.        , 0.14953944],
       [0.14953944, 1.        ]])
```

```python
from scipy.stats import pearsonr

x = [7, 18, 29, 2, 10, 9, 9]
y = [1, 6, 12, 8, 6, 21, 10]
pearsonr(x, y)
```

```
(0.14953944411984524, 0.74896289035078)
```

```python
def find_corr_x_y(x,y):
    n = len(x)
    prod = []
    for xi,yi in zip(x,y):
        prod.append(xi*yi)

    sum_prod_x_y = sum(prod)

    sum_x = sum(x)
    sum_y = sum(y)

    squared_sum_x = sum_x**2
    squared_sum_y = sum_y**2

    x_square = []
    for xi in x:
        x_square.append(xi**2)
    x_square_sum = sum(x_square)

    y_square=[]
    for yi in y:
        y_square.append(yi**2)
    y_square_sum = sum(y_square)

    # Use formula to calculate correlation
    numerator = n*sum_prod_x_y - sum_x*sum_y
    denominator_term1 = n*x_square_sum - squared_sum_x
    denominator_term2 = n*y_square_sum - squared_sum_y
    denominator = (denominator_term1*denominator_term2)**0.5
    correlation = numerator/denominator

    return correlation
```

Outline

- ➢ Mean, Variance and Standard Deviation: Review
- ➢ What is Correlation?
- ➢ What is Covariance?
- ➢ What is Correlation Coefficient
- ➢ Template Matching
- ➢ Summary

Application to Patch Matching

P_1 P_2 P_3 P_4

$\rho_{P_1 P_2} = 0.55$

$\rho_{P_1 P_3} = 0.23$ ➡ Ảnh P_2 giống với ảnh P_1 hơn so với P_3 và P_4

$\rho_{P_1 P_4} = 0.30$

P_1 $P_2 = P_1 + 50$ $P_3 = 1.2P_1 + 10$

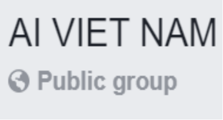
$\rho_{P_1 P_2} = 0.9970$

$\rho_{P_1 P_3} = 0.9979$ ➡ ρ hoạt động tốt dưới sự thay đổi tuyến tính

```python
# aivietnam.ai

import numpy as np
from PIL import Image

# load ảnh và chuyển về kiểu list
image1 = Image.open('images/img1.png')
image2 = Image.open('images/img2.png')
image3 = Image.open('images/img3.png')
image4 = Image.open('images/img4.png')

image1_list = np.asarray(image1).flatten().tolist()
image2_list = np.asarray(image2).flatten().tolist()
image3_list = np.asarray(image3).flatten().tolist()
image4_list = np.asarray(image4).flatten().tolist()


# tính correlation coefficient
corr_1_2 = find_corr_x_y(image1_list, image2_list)
corr_1_3 = find_corr_x_y(image1_list, image3_list)
corr_1_4 = find_corr_x_y(image1_list, image4_list)

print('corr_1_2:', corr_1_2)
print('corr_1_3:', corr_1_3)
print('corr_1_4:', corr_1_4)
```

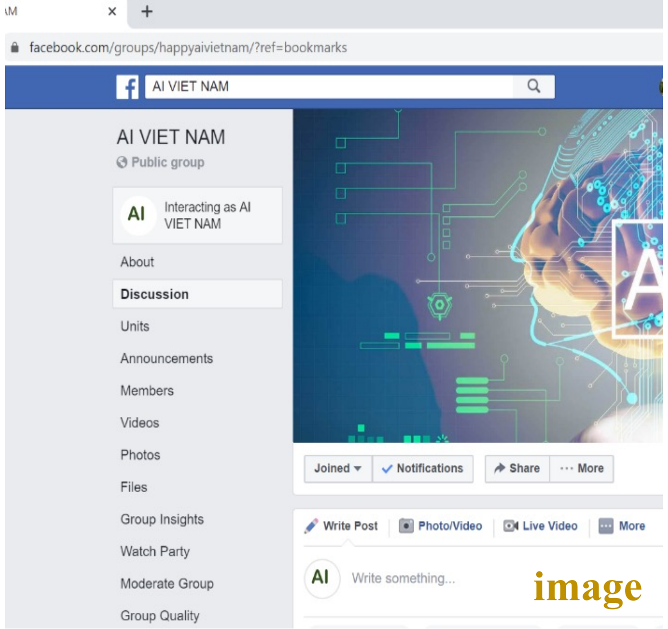
Correlation Coefficient

❖ Application to template matching



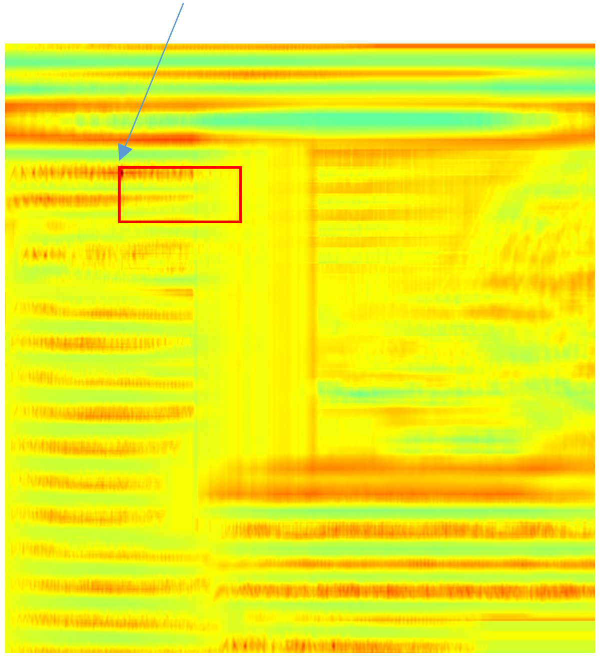
template

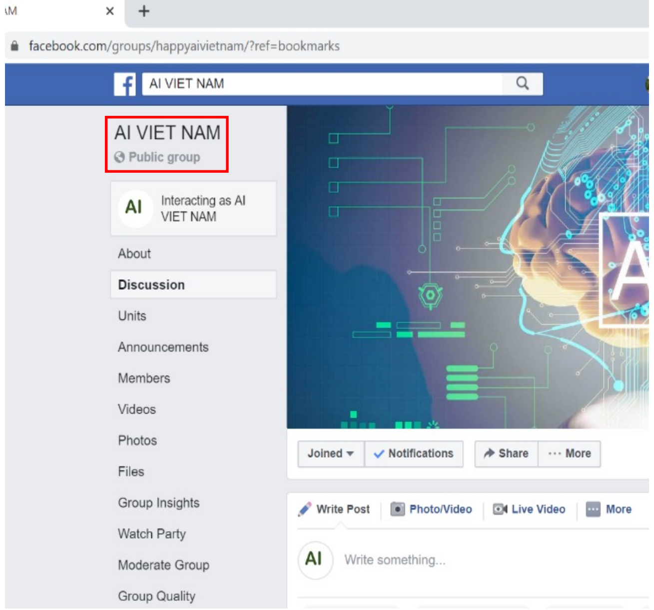
Tìm template có trong hình image



image

max

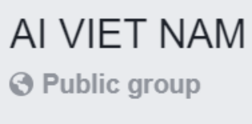


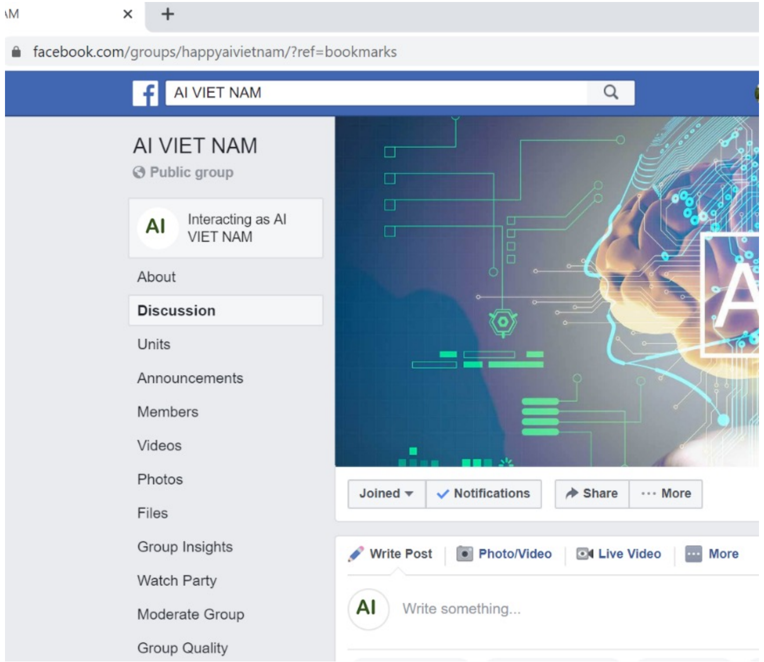


Output

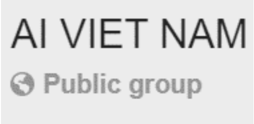
Correlation Coefficient

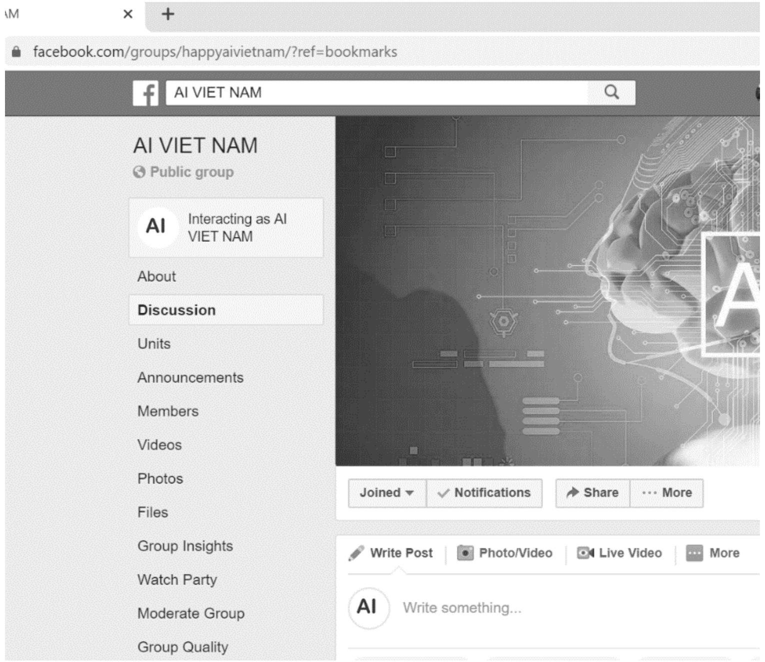
❖ Application to template matching





Color images





Grayscale images

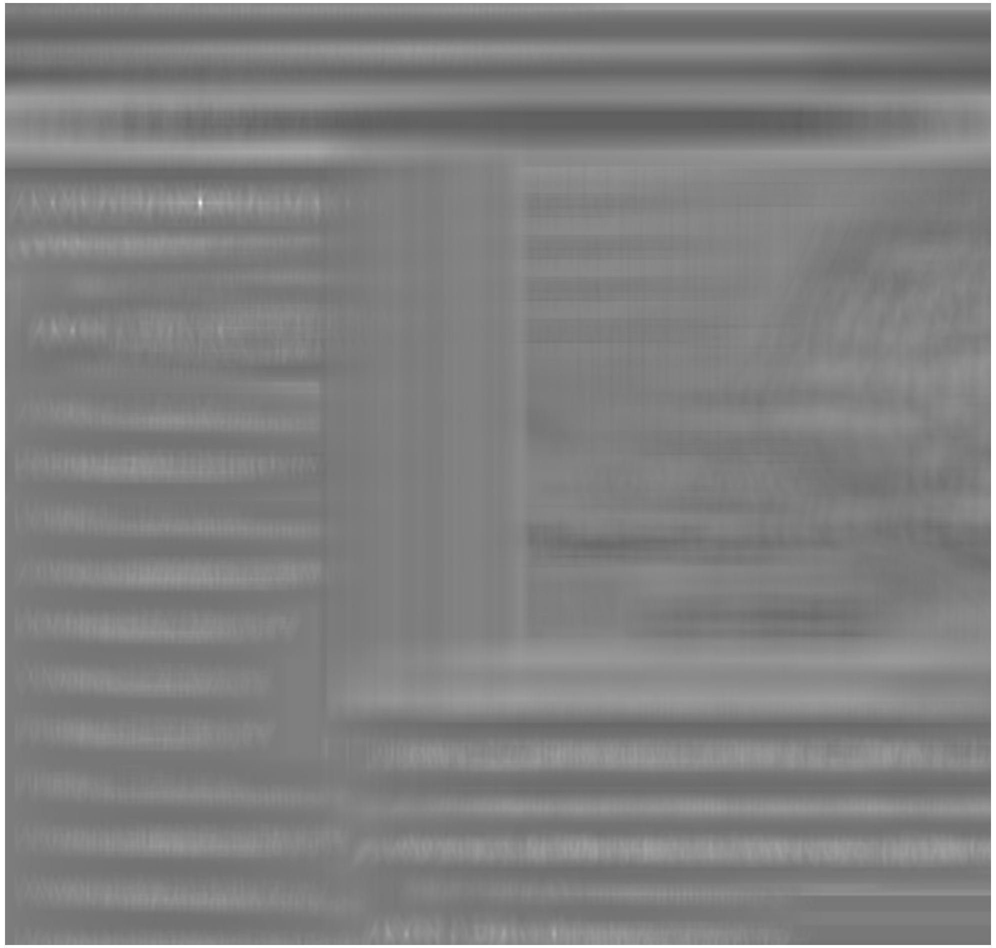
```
grayscale_image = cv2.cvtColor(color_image, cv2.COLOR_BGR2GRAY)
output = cv2.matchTemplate(image, template, cv2.TM_CCOEFF_NORMED)
```

AI VIET NAM
@aivietnam.edu.vn

Correlation Coefficient

❖ Application to template matching

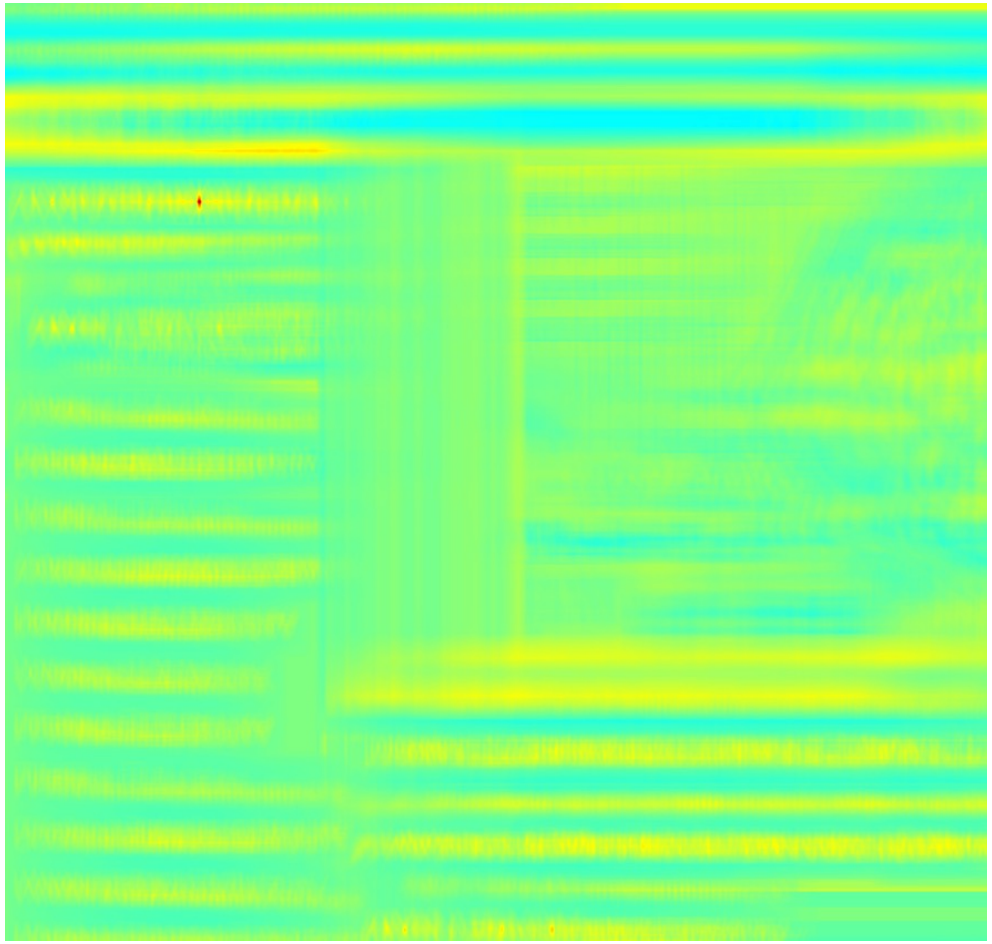
```python
# template matching

import cv2
import numpy as np
from matplotlib import pyplot as plt

# Load image and convert to grayscale
image = cv.imread('image.png',1)
gray  = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print(type(gray[0][0]))

template = cv.imread('template.png',0)
w, h = template.shape[::-1]

# Apply template Matching
corr_map = cv2.matchTemplate(gray, template, cv2.TM_CCOEFF_NORMED)

# save correlation map
corr_map = (corr_map+1.0)*127.5
corr_map = corr_map.astype('uint8')
cv.imwrite('corr_map_grayscale.png', corr_map)
```



Correlation Map

Correlation Coefficient

❖ Application to template matching



```python
import cv2
import numpy as np
from matplotlib import pyplot as plt

# Load image and convert to grayscale
image = cv.imread('image.png',1)
gray  = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

template = cv.imread('template.png',0)
w, h = template.shape[::-1]

# Apply template Matching
corr_map = cv2.matchTemplate(gray, template, cv2.TM_CCOEFF_NORMED)
min_val, max_val, min_loc, max_loc = cv.minMaxLoc(corr_map)

# take minimum
top_left = max_loc
bottom_right = (top_left[0] + w, top_left[1] + h)

# draw
cv.rectangle(image, top_left, bottom_right, (0, 255, 0), 2)

# save corr map in grayscale
corr_map = (corr_map+1.0)*127.5
corr_map = corr_map.astype('uint8')
cv.imwrite('corr_map_grayscale.png', corr_map)

# applyColorMap
corr_map = cv2.applyColorMap(corr_map, cv2.COLORMAP_JET)

# save results
cv.imwrite('corr_map_color.png', corr_map)
cv.imwrite('result.png', image)
```

Outline

- ➢ Mean, Variance and Standard Deviation: Review
- ➢ What is Correlation?
- ➢ What is Covariance?
- ➢ What is Correlation Coefficient
- ➢ Template Matching
- ➢ Summary